

RESEARCH ARTICLE SUMMARY

CELL BIOLOGY

Myelin is repaired by constitutive differentiation of oligodendrocyte progenitors

Yevgeniya A. Mironova, Brendan Dang, Dongeun Heo, Yu Kang T. Xu, Angela Yu-Huey Hsu, Jaime Eugenin von Bernhardt, Gian Carlo Molina-Castro, Anya A. Kim, Jing-Ping Lin, Daniel S. Reich, Dwight E. Bergles*



Full article and list of author affiliations:
<https://doi.org/10.1126/science.adu2896>

INTRODUCTION: Insulating sheaths of myelin around nerve axons allow rapid signal propagation in the mammalian central nervous system (CNS) that is crucial for processing sensory information, coordinating movement, and enabling higher-order cognitive functions. In the brain, myelin is exclusively produced by oligodendrocytes that are formed through differentiation of resident progenitor cells. These oligodendrocyte precursor cells (OPCs) remain abundant and widely distributed in the CNS throughout life, enabling prolonged production of new myelin to establish, modify, and repair damage to oligodendrocytes that results from trauma or disease. However, the mechanisms that control differentiation of these endogenous progenitors are not well understood.

RATIONALE: Understanding the differentiation potential of OPCs and the effects of environmental changes such as brain aging and inflammation on the process of differentiation may help develop new strategies for repairing and replacing myelin. However, tracking OPC differentiation in the CNS is challenging, because most OPCs that initiate lineage progression die before they mature into myelin-producing oligodendrocytes. Thus, recording the emergence of new oligodendrocytes alone underestimates the extent of OPC differentiation. OPC differentiation is also frequently inferred by monitoring their proliferation, as they maintain a constant density throughout life by replacing cells that attempt to differentiate; however, OPCs can directly differentiate into oligodendrocytes without proliferation and are vulnerable to various insults, which confound the interpretation of proliferation as an indirect indication of differentiation.

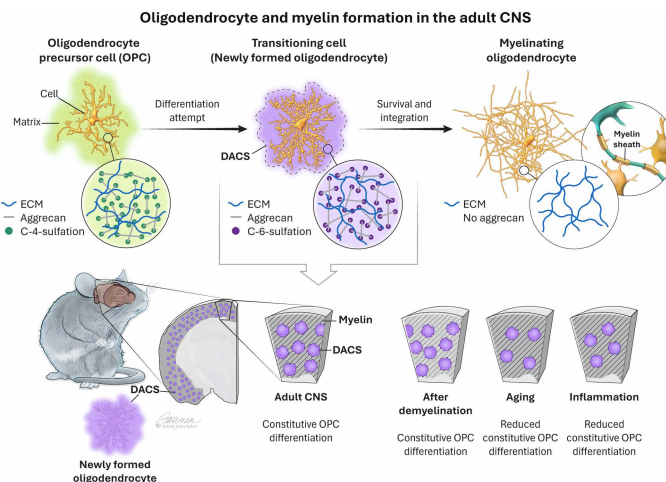
RESULTS: Using transcriptional profiling in mouse, marmoset, and human, we found that OPCs exhibit a characteristic change in gene

expression during the initial stages of differentiation that selectively modifies the surrounding extracellular matrix (ECM). This modified ECM persists for up to 10 days, allowing detection of OPC differentiation attempts, even when they are unsuccessful and result in death and removal of the transitioning cell. Leveraging this new insight provided improved resolution of oligodendrogenesis in the brain. We used immunostaining and in vivo longitudinal time-lapse imaging over weeks in the adult mouse brain to define the spatial and temporal dynamics of OPC lineage progression and assessed the effects of aging, demyelination, and inflammation on this process. These studies revealed that OPC differentiation was not correlated with the eventual pattern of oligodendrocytes achieved, occurred in areas devoid of oligodendrocytes and myelin, and was notably insensitive to oligodendrocyte loss and demyelination.

CONCLUSION: The generation of new oligodendrocytes in the adult brain is governed predominantly by widespread, constitutive lineage progression of OPCs that is uncoupled from myelin demand. This process appears to be an extension of their behavior in the developing CNS, designed to extend the period of myelination rather than replace myelin lost through injury or disease. The production of new oligodendrocytes is therefore constrained by this occasional, constitutive lineage progression of OPCs, which declines with aging and is further suppressed by inflammation and homeostatic proliferation of these progenitors when they are damaged. Defining the factors that influence this constitutive behavior of OPCs may provide additional means to accelerate lineage progression of these ubiquitous progenitors to enhance remyelination in diseases such as multiple sclerosis. □

*Corresponding author. Email: dbergles@jhmi.edu Cite this article as Y. A. Mironova *et al.*, *Science* 391, eadu2896 (2026). DOI: [10.1126/science.adu2896](https://doi.org/10.1126/science.adu2896)

ECM modification allows tracking of OPC differentiation attempts. OPCs alter the sulfation pattern on aggrecan as they differentiate. This change in aggrecan sulfation is recognized by the CS56 antibody, revealing immunoreactive dandelion clock-like structures (DACS). Because this modified aggrecan in the ECM persists whether differentiation is successful or fails, DACS reveal each OPC differentiation attempt. Retrospective analysis of DACS revealed that extensive demyelination does not alter OPC differentiation, but that these transitions decline with aging and brain inflammation.



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Myelin is repaired by constitutive differentiation of oligodendrocyte progenitors

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Oligodendrocytes form myelin sheaths around axons to enable rapid signaling within neural circuits. The generation of new oligodendrocytes through differentiation of oligodendrocyte precursor cells (OPCs) promotes myelin plasticity and repair in the adult brain. Here, we performed genetic interrogation and in vivo analysis of OPCs in the mouse brain to determine their differentiation dynamics. Our results show that OPCs attempt to differentiate throughout the adult central nervous system with spatial and temporal regularity. The differentiation rate was not influenced by myelin demand or oligodendrocyte loss and declined with age and in response to acute inflammation. The results suggest that OPC differentiation is governed primarily by constitutive processes and might be negatively influenced by aging and inflammation.

The adult central nervous system (CNS) has a limited capacity for repair. As a result, loss of neurons through injury or disease often leads to persistent functional deficits (1, 2). However, supportive glial cells in the CNS can be rapidly replaced to maintain homeostasis (3–6). The most prominent regeneration occurs in response to destruction of oligodendrocytes and their myelin sheaths (7–9), which allow rapid conduction of nerve impulses, regulate neuronal excitability, and provide metabolic support to underlying axons (10–12). The generation of new oligodendrocytes continues into adulthood (13–20), can be altered by behavioral experience (15, 21–23), and is crucial to restore myelin in diseases such as multiple sclerosis (MS) (24, 25). This ongoing oligodendrogenesis is enabled by the differentiation of oligodendrocyte precursor cells (OPCs), which remain abundant and widely distributed in both gray and white matter throughout life (26–28). Present models of myelin repair suggest that these progenitors are induced to differentiate in response to changes in neural activity and detect local oligodendrocyte loss to regenerate myelin (29–32). However, it has been difficult to track OPC differentiation events in vivo, as most OPCs that begin to transform into oligodendrocytes in the adult CNS die at the premyelinating stage and are rapidly cleared (15, 16, 33–35). As a result, little is known about the mechanisms that regulate OPC lineage progression in the mature brain. Uncovering the basis of this differentiation process may provide the means to reassert proper myelination programs to reduce developmental disability and enhance the recovery of myelin lost through injury, normal aging, or disease. Here, we explored the molecular changes associated with the

initial stages of OPC differentiation and examined the differentiation of these endogenous progenitors in the adult brain to define how development, demyelination, aging, and inflammation affect their lineage progression.

OPCs produce extracellular matrix proteins

Oligodendrogenesis—the de novo generation of new oligodendrocytes—is required to establish myelin sheaths around axons in the CNS, as existing oligodendrocytes, once mature, exhibit a limited ability to produce new sheaths (13, 15, 36). This process begins with OPC lineage progression and is associated with profound changes in gene expression that guide their morphological and functional transformation (fig. S1A). Once initiated, these transforming cells either die or fully mature into myelin-forming oligodendrocytes. Histological analyses and in vivo imaging studies indicate that oligodendrogenesis is highly inefficient in both the developing and adult CNS, with most differentiating cells undergoing apoptosis before myelin sheath formation (15, 16, 33–35, 37). Because most cells that begin lineage progression die, it has been difficult to assess the extent of differentiation brain-wide in different physiological and pathological conditions. To establish a new strategy for in vivo assessment of OPC lineage progression, we interrogated single-cell RNA sequencing (scRNA-seq) profiles of oligodendroglia isolated from the adult CNS to define the transcriptional changes associated with this transition (38, 39). The most highly expressed mRNAs in OPCs included chondroitin sulfate proteoglycans (CSPGs), such as those encoded by *Cspg4*, *Vcan*, *Ptprz1*, *Bcan*, and *Acan*, many of which were enriched in OPCs relative to other major CNS cell types (astrocytes, microglia, and neurons) (fig. S1B). This expression pattern was conserved between mouse, marmoset, and human (38–41) (fig. S1C). CSPG expression was particularly high in OPCs and newly formed oligodendrocytes (NFOs) in the initial stages of differentiation. These transcripts decreased to near undetectable amounts in mature oligodendrocytes, suggesting that changes in CSPG expression could be used to identify oligodendroglia in the early stages of differentiation. To further define state- and age-specific dynamics of CSPG expression by OPCs, we queried a scRNA-seq database of OPCs isolated from the mouse cerebral cortex (fig. S1, D to E) (42). This analysis confirmed that OPCs rapidly down-regulate CSPG expression in the initial stages of differentiation (fig. S1F and table S1). Moreover, this pattern of CSPG expression by OPCs persisted from the young [post-natal day 30 (P30)] to the aged (P720) CNS, indicating that it can be used to monitor OPC lineage progression across the life span (fig. S1G and table S2).

OPCs exhibit a distinctive pattern of CSPG production in the brain

To analyze CSPG production by OPCs in situ, we performed immunostaining of mouse brains with an anti-chondroitin sulfate (CS) antibody (CS56), which revealed that the adult brain contains discrete rosettes of CS immunoreactivity of variable intensity that are widely distributed throughout the brain (Fig. 1A and fig. S2). These CS56 immunoreactive (+) intensities have been described in brain tissue of multiple species (43–45) and have been termed dandelion clock-like structures (DACS), which were thought to reflect extracellular matrix (ECM) production by astrocytes (43, 45, 46). To determine whether this labeling is specific for CSPGs and to monitor their production, we generated transgenic mice in which the bacterial enzyme chondroitinase ABC (ChABC), which selectively digests CSPG carbohydrate chains (47), can be inducibly expressed (see Materials and methods and fig. S12A). This approach avoids tissue injury, reactive gliosis, and variable enzymatic digestion associated with direct ChABC injection (48). In these mice, a mammalian-optimized *ChABC* coding sequence (48) tagged with a triple hemagglutinin (3xHA) tag is expressed under the control of the tetracycline inducible promoter pTRE-Tight (*ChaseR-TRE*) (Fig. 1B). *ChaseR-TRE* mice were crossed with mice that express

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Fig. 1. Enriched CSPG expression is associated with differentiating OPCs.

(A) CS56 immunostaining of CSPGs in a sagittal section of a 3- to 4-month-old C57BL/6 mouse. A dashed box indicates the somatosensory cortex area shown in the higher-magnification panel to the right.

(B) A genetic and experimental strategy of DOX-inducible expression of mammalian-optimized ChABC in the murine CNS (*ChaseR-TRE;CAMKII-tTA*). A transgenic sequence consists of a tTA-regulated *pTRE-Tight* promoter, a ChABC coding sequence in frame with a 3xHA tag sequence, a woodchuck posttranscriptional response element (WPRE), and bovine growth hormone polyA sequence (pA). *tTA* is expressed under the control of a *CAMKII* promoter (*CAMKII-tTA*).

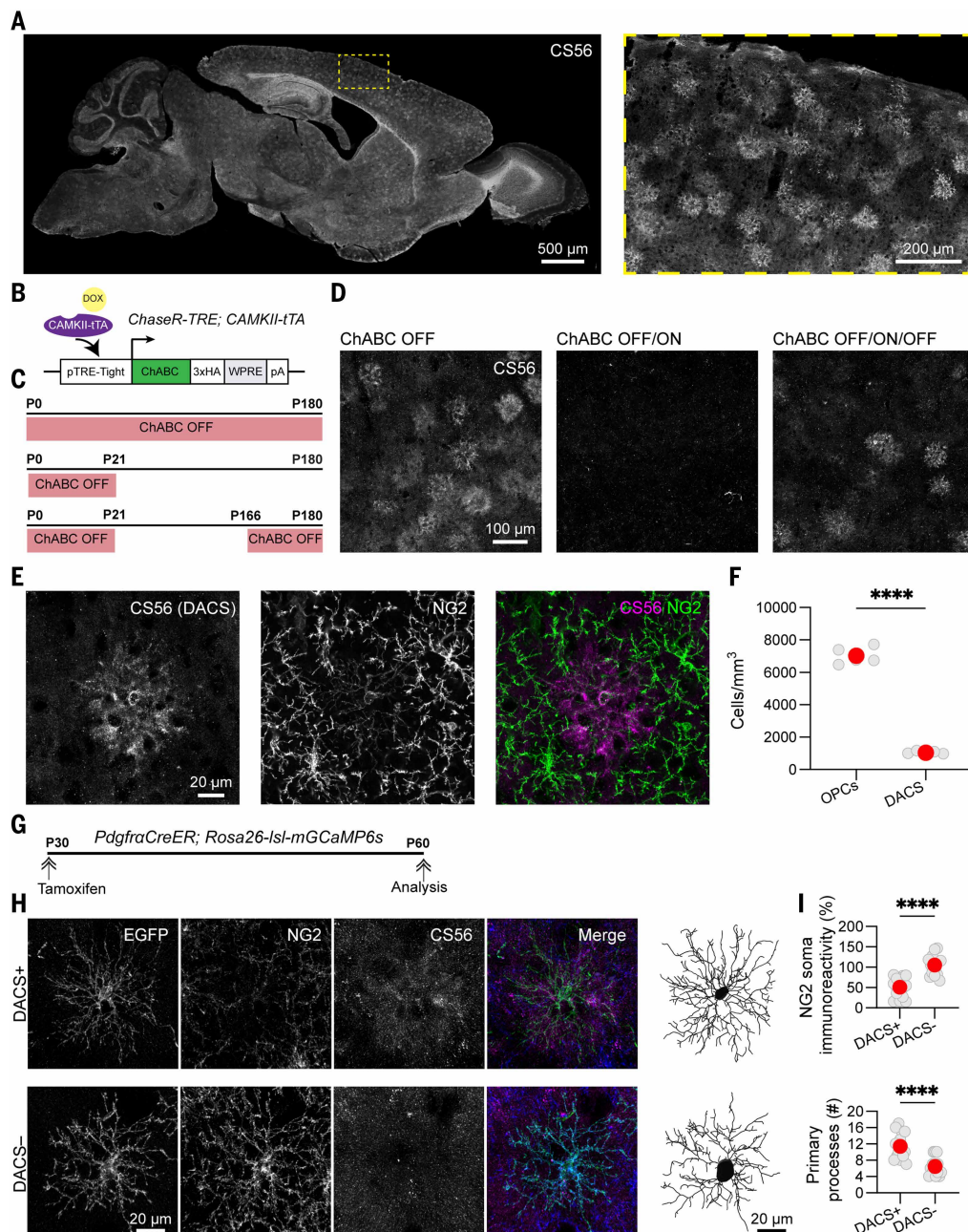
(C) Experimental strategy of DOX-regulated ChABC expression. DOX was administered to *ChaseR-TRE;CAMKII-tTA* mice continuously to prevent ChABC expression (ChABC OFF; top), removed at P21 to induce ChABC expression (ChABC ON; middle), and removed at P21 and reintroduced at 14 days before tissue collection (ChABC OFF/ON; bottom).

(D) CS56 immunostaining of CSPGs in the somatosensory cortex of *ChaseR-TRE;CAMKII-tTA* mice with experimental strategies outlined in (C). (E) A high-magnification image of immunostained CS56 CSPGs (left) and NG2 OPCs (middle) in the somatosensory cortex, in the same tissue as in (A). A high-signal intensity cluster of CS56 is referred to as a DACS. The merged panel (right) demonstrates DACS colocalization with an NG2⁺ OPC. (F) Quantification of NG2⁺ OPC and DACS densities in the somatosensory cortex of 3- to 4-month-old C57BL/6 mice. Filled circles and error bars represent means $\pm$ SEM. Data are as follows: OPCs, 7022 ± 231 cells/mm³, $n = 5$ mice; DACS, 1040 ± 36 DACS/mm³, $n = 5$ mice; $p < 0.0001$, unpaired two-tailed Student's *t* test.

(G) Experimental strategy for inducing sparse recombination of membrane anchored GFP in the oligodendrocyte (OL) lineage of *Pdgfra-CreER; Rosa26-IsI-mGCaMP6s* mice. A single intraperitoneal injection of tamoxifen was given at P30, and tissue was collected and analyzed at P60. (H) Representative images of immunolabeled DACS⁺ and DACS⁻ EGFP⁺ NG2⁺ cells. The graphics on the right show filament reconstructions of individual EGFP⁺ cells. (I) Quantification of soma NG2 immunoreactivity (top). Mean fluorescence intensity of the soma of analyzed cells was normalized to the same value of a nearest NG2⁺ DACS⁻ OPC. Filled circles and error bars represent means $\pm$ SEM. Data are as follows: DACS⁺ cells, $51 \pm 6\%$, $n = 15$ cells from four mice; DACS⁻ cells, $105 \pm 7\%$, $n = 14$ cells from four mice; $p < 0.0001$, Mann-Whitney test. Analysis of data averages per animal is as follows: DACS⁺, $51 \pm 2\%$; DACS⁻, $106 \pm 2\%$; $n = 4$ mice; $p = 0.0286$, Mann-Whitney test. Quantification of a number of primary processes of EGFP⁺ cells is shown at the bottom. Data are as follows: DACS⁺ cells, 11 ± 1 processes per cell, $n = 14$ cells from four mice; DACS⁻ cells, 7 ± 1 processes per cell, $n = 15$ cells from four mice; $p < 0.0001$, Mann-Whitney test. Analysis of data averages per animal is as follows: DACS⁺, 11 ± 1 processes per cell; DACS⁻, 6 ± 1 processes per cell; $n = 4$ mice; $p = 0.0286$, Mann-Whitney test. **** $p < 0.0001$.

tetracycline-controlled transactivator (tTA) under control of a forebrain-specific neuronal promoter (49) (*ChaseR-TRE;CAMKII-tTA*). When *ChaseR-TRE;CAMKII-tTA* mice were chronically exposed to doxycycline (DOX), HA immunoreactivity was absent and DACS and wisteria floribunda agglutinin (WFA) labeling that detects CSPG-containing perineuronal nets (PNNs) were visible, indicating effective suppression of transgene expression (Fig. 1, C and D, and fig. S3). Upon DOX withdrawal at weaning, HA⁺ neurons were visible in the forebrain of adult

mice, and CS56⁺ DACS and WFA⁺ PNNs were absent, indicating efficient reinduction of ChABC expression (Fig. 1, C and D, and fig. S3). ECM components are among the most long-lived proteins in the CNS (50–52). To determine whether these CSPG-rich DACS are dynamic structures, DOX was reintroduced for 14 days after several months without DOX to allow newly synthesized CSPGs to accumulate. In these mice, no HA⁺ neurons were visible, but DACS and PNNs were present, indicating that DACS are continuously formed in the CNS (Fig. 1,



C and D, and fig. S3). To determine whether these CSPG accumulations are associated with OPCs, we coimmunolabeled CS56 with NG2, a membrane-anchored CSPG that is highly expressed by OPCs (53, 54). A subset of NG2⁺ OPCs in the somatosensory cortex exhibited CS56 immunoreactivity around their soma and processes, consistent with the transcriptional evidence of CSPG production by OPCs (Fig. 1E). To determine whether DACs are exclusively produced by OPCs, we generated a transgenic mouse line that enables selective ablation of OPCs (*OPC-Terminator*^{fl/+}). A diphtheria toxin receptor (DTR) genomic sequence was inserted in a *Cspg4* (NG2) intron and, upon *Cre* recombination, is expressed under the control of the *Cspg4* promoter (see Materials and methods and figs. S4A and S12B). A combination of *Olig2* and *Cspg4* promoters of *Cre* and *Terminator* restricted DTR expression to OPCs. A single intraperitoneal injection of diphtheria toxin (DT) was sufficient to ablate nearly all cortical OPCs within 24 hours, as assessed using in vivo two-photon imaging in the motor cortex of *Olig2-Cre;Matn4-mEGFP;OPC-Terminator*^{fl/+} mice (fig. S4, B and C, and movie S1). Post hoc NG2 immunolabeling of tissue isolated from these mice revealed that OPCs repopulated the cortex as soon as 8 days after DT administration, consistent with their rapid homeostatic response to cell loss (55, 56); however, almost no DACs were detected in the same cortical region at this time point, supporting the conclusion that DACs are exclusively produced by OPCs (fig. S4, D and E).

DACS surround differentiating OPCs

Although DACs are produced by OPCs, they are far less abundant, suggesting that they may be associated with a particular state of these cells (Fig. 1F). Close inspection of CS56-immunolabeled sections revealed that DACS⁺ OPCs exhibited weaker NG2 immunoreactivity, a feature of oligodendroglia in the initial stages of differentiation (Fig. 1E) (57). OPC lineage progression is associated with increased process formation and extension into the surrounding neuropil as they search for receptive axons to myelinate (57, 58). To define the morphological features of DACS-associated OPCs, we genetically labeled individual OPCs with membrane-anchored green fluorescent protein (GFP) by administering a low dose of tamoxifen to *Pdgfra-CreER;Rosa26-lsl-mGCaMP6s* mice (17, 59) (Fig. 1G). EGFP⁺ DACS⁺ (where EGFP stands for enhanced GFP) cells had lower NG2 immunoreactivity and more primary processes compared with EGFP⁺ cells devoid of surrounding DACs (Fig. 1, H and I). Together, these results indicate that OPCs induce a transient change in the surrounding ECM as they differentiate, giving rise to DACs.

Transformation of OPCs into oligodendrocytes requires ~2 weeks in vivo (57), with transitioning cells exhibiting distinct morphological, physiological, and transcriptional features at different stages of maturation. To determine how long DACs remain associated with differentiating cells, we genetically tagged OPCs by administering a saturating dose of tamoxifen to *Pdgfra-CreER;Rosa26-lsl-EGFP (RCE)* mice (60), inducing EGFP expression in OPCs that persists as they move through different stages of the differentiation process (Fig. 2A). Quantitative analysis of DACs in the somatosensory cortex revealed that >50% of DACs colocalized with EGFP⁺ NG2⁺ cells (OPCs) and EGFP⁺ NG2⁻ cells (NFOs and myelinating oligodendrocytes) (Fig. 2, B and C). All EGFP⁺ DACS⁺ OPCs colocalized with BCAS1, a gene up-regulated at the NFO stage (61) (Fig. 2, B and C). In accordance with the decline in CSPG production with differentiation, the vast majority of DACs did not colocalize with mature oligodendrocytes (ASPA⁺) (Fig. 2, B and C, and fig. S1). Colocalization of DACs with oligodendrocyte lineage cells was also observed in *Olig2-Cre;RCE* mice, providing further evidence that this matrix modification is induced by differentiating OPCs (fig. S5). Previous in vivo imaging studies revealed that most differentiating cells in the cortex fail to integrate as mature oligodendrocytes (15). Consistent with this observation, DACs were also observed surrounding dying NFOs (arrowheads in fig. S6), characterized by atrophied EGFP⁺ processes and condensed somata, suggesting that EGFP⁺ NG2⁻ DACs

(not associated with OPCs or NFOs) reflect temporary persistence of modified ECM at sites where NFOs failed to survive (Fig. 2, B and C). Together, these results indicate that DACs most likely mark the site of each OPC differentiation attempt.

OPCs in the CNS exhibit robust homeostasis, in which differentiation is associated with rapid proliferation of a neighboring OPC to maintain their density (57). If DACs reflect oligodendrogenesis, they should be associated with nearby OPC proliferation. To assess this proliferation of OPCs relative to DACs, we ended DOX administration in *ChaseR-TRE;CAMKII-tTA* mice to mark newly modified ECM and then administered 5-ethynyl-2'-deoxyuridine (EdU) for 14 days (62, 63) (Fig. 2D). In these mice, all 26 DACs analyzed from three mice were located next to EdU⁺ OPCs, and EdU⁺ OPCs were closer to DACs than predicted by random chance (Fig. 2, E and F; see Materials and methods for a detailed description of analysis). These results are consistent with the hypothesis that OPCs modify CSPGs in the surrounding ECM as they differentiate and that most, if not all, DACs are associated with differentiating oligodendroglia.

Previously published time-lapse imaging in the adult CNS demonstrated that it takes ~10 days for successful differentiation of OPCs into myelin sheath-producing oligodendrocytes (57). If DACs provide a snapshot of all differentiation attempts that occurred in the past ~10 days, the number of DACs should match the number of OPCs that initiate lineage progression detected through longitudinal in vivo imaging over a 10-day period. To test this hypothesis, we assessed OPCs undergoing these transitions through time-lapse, two-photon volumetric imaging in the adult somatosensory cortex of *Pdgfra-CreER;RCE* mice and compared the number of differentiation attempts with the number of DACs within that region in mice of the same age and genotype (fig. S7, A to C, and movies S2 and S3). When normalized to the total number of OPCs in the imaging volume, we found close correspondence between OPC differentiation attempts and the number of DACs (fig. S7, D to F). These results support the conclusion that ECM modification by OPCs enables detection of all differentiation attempts, regardless of success. As this change in ECM persists beyond the lifetime of cells that fail to differentiate, DACs provide a powerful means to assess the extent of OPC differentiation without genetic labeling or other manipulations.

DACS represent changes in aggrecan sulfation

Transcriptional analyses suggest that OPCs contribute numerous CSPGs to the ECM (fig. S1). Of these, aggrecan carries, by far, the most CS chains (>100) (64) (Fig. 3A) and is highly abundant in the cartilage-rich tissue used to generate the CS56 antibody (65). Thus, despite its relatively low expression, aggrecan may contribute most of the CS recognized by CS56. Consistent with this hypothesis, DACs were abolished when *Acan*, the gene that encodes aggrecan, was selectively deleted from OPCs (*Pdgfra-CreER;Acan*^{fl/fl} mice; hereafter “*Acan* cKO” mice) (66) (Fig. 3, B and C). However, the presence of *Acan* mRNA in OPCs before differentiation, and the selective emergence of CS56 immunoreactivity at the onset of differentiation, suggests that DACs are unlikely to represent simply enhanced expression of aggrecan. CSPGs are extensively posttranslationally modified through the addition of sulfur atoms at different positions on CS chains, creating distinct sulfation patterns (67) (Fig. 3D and fig. S1). Transcriptional analysis suggests that OPCs in the initial stages of differentiation undergo a shift in expression of genes that modify sulfur addition to aggrecan. Expression of chondroitin 4 sulfotransferase, CS4T (*ChstII*), which catalyzes sulfur addition to CSPGs at the C-4 position of *N*-acetyl-D-galactosamine (GalNAc) (68, 69), declines as OPCs differentiate in concert with an increase in expression of arylsulfatase B (*Arsb*), a hydrolase that removes sulfur atoms from the C4 position (Fig. 3E) (70). Coincident with these changes, expression of chondroitin 6 sulfotransferase CS6T (*Chst3*), which catalyzes sulfur addition at the C-6 position of GalNAc (71), transiently increases in NFOs, before

Fig. 2. DACS indicate OPC differentiation attempts.

(A) Experimental strategy for OPC lineage tracing in *Pdgfra-CreER;RCE* transgenic mice. Tamoxifen was administered daily for five consecutive days at P30, and tissue was collected and analyzed at P60.

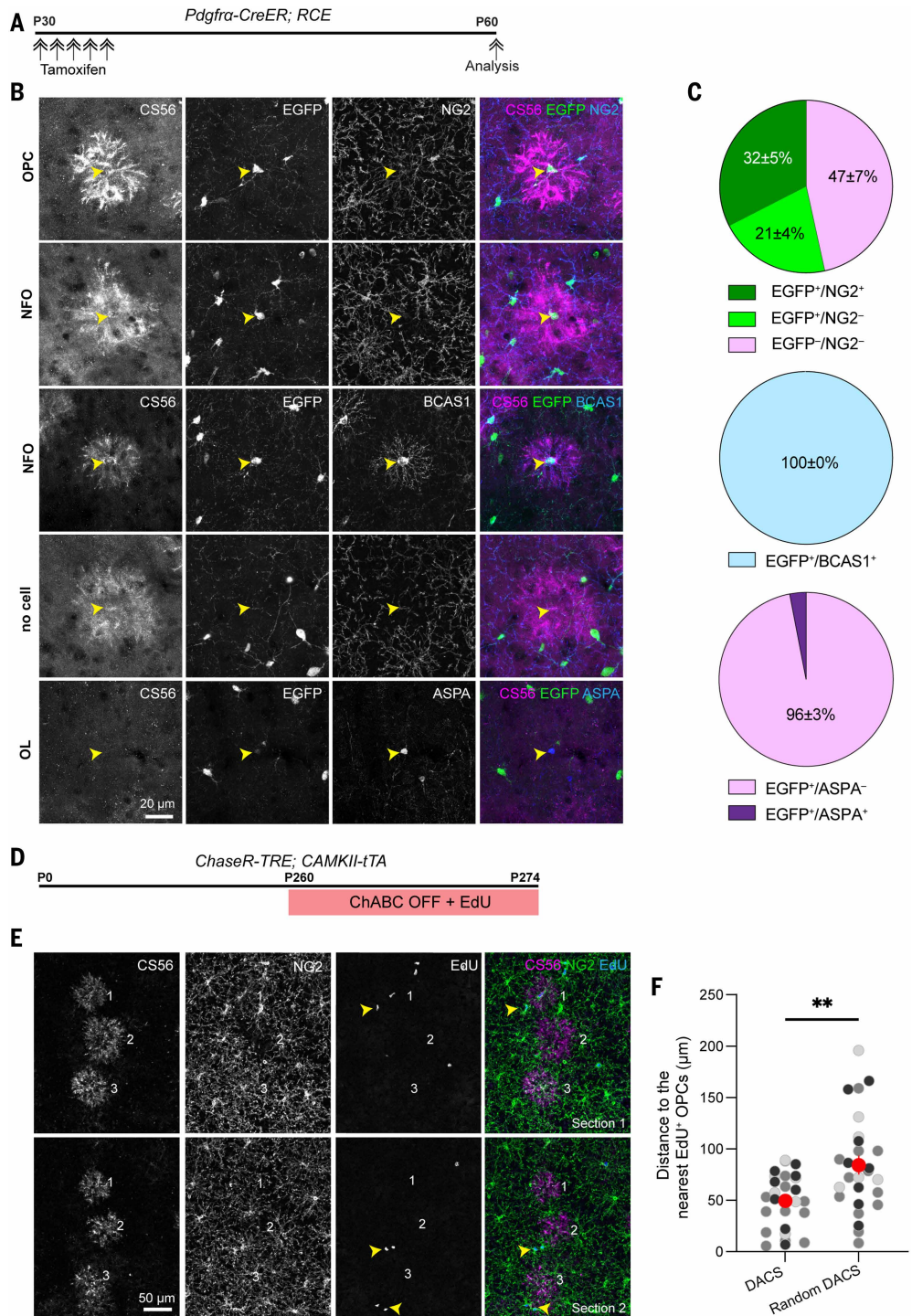
(B) Representative images of immunolabeled GFP⁺ CS56⁺ cells in the cerebral cortex of *Pdgfra-CreER;RCE* mice. Triple labeled CS56⁺ EGFP⁺ NG2⁺ cells were classified as “OPC”; CS56⁺ EGFP⁺ NG2⁻ and CS56⁺ EGFP⁻ NG2⁻ cells as “NFO”; CS56⁻ EGFP⁻ NG2⁻ cells as “no cell”; and EGFP⁺ ASPA⁺ cells as “OL.” Yellow arrowheads indicate DACS centroids.

(C) Quantification of the proportion of DACS colocalized with EGFP-labeled OL lineage cells (top); quantification of the proportion of DACS colocalized with EGFP⁻ and BCAS1-labeled cells (middle); and quantification of the proportion of DACS colocalized with EGFP⁻ and ASPA-labeled cells (bottom). Values listed are means $\pm$ SEM, and $n = 5$ mice (for the ASPA analysis, $n = 4$ mice).

(D) Experimental strategy for simultaneous labeling of recently formed DACS and proliferating OPCs in *ChaseR-TRE;CAMKII-tTA* mice. The pink bar denotes continuous administration of DOX and EdU.

(E) Representative images of DACS, EdU, and NG2 immunolabeling in adjacent sections that span the volume of individual DACS and neighboring OPCs. Numbers label individual DACS across adjacent sections, and yellow arrowheads indicate nearest neighbor EdU⁺ NG2⁺ OPCs.

(F) Quantification of distances between DACS (centroids) and nearest EdU⁺ OPCs as well as distances between randomly distributed DACS (centroids) and nearest EdU⁺ OPCs. Filled red circles represent means $\pm$ SEM, and shading is used to distinguish values collected from different mice. Data are as follows: DACS, $50 \pm 2 \mu\text{m}$; Random DACS, $84 \pm 10 \mu\text{m}$; $n = 26$ cells from three mice for both groups; $p = 0.0020$, unpaired, two-tailed Student's t test. Analysis of data averages per animal is as follows: DACS, $52 \pm 6 \mu\text{m}$; Random DACS, $88 \pm 6 \mu\text{m}$; $n = 3$ mice; $p = 0.0439$, unpaired, two-tailed Student's t test. $**p < 0.01$.



decreasing in mature oligodendrocytes (Fig. 3E and table S3) (38, 42). These results suggest that OPCs may acutely change the sulfation pattern on aggrecan from the C-4 to the C-6 position as they differentiate. If DACS represent 6-O-sulfated aggrecan, then altering the expression of sulfotransferases should influence the pattern of CS56 immunoreactivity. Indeed, conditional knockout (deletion) of *Chst3* within OPCs using conditional *Chst3* knockout mice (*Pdgfra-CreER;Chst3^{fl/fl}*; hereafter “*Chst3* cKO” mice; see Materials and methods) abolished DACS (Fig. 3, F to H, and fig. S12C). These data indicate that OPCs acutely modify the sulfation pattern on aggrecan as they begin to

differentiate, and this change is detected as DACS through CS56 immunostaining.

ECM secretion by OPCs modulates their homeostasis in vivo

To assess the functional role of this matrix modification by OPCs, we performed in vivo two-photon imaging in control and *Chst3* cKO mice during periods of active (P75+) and reduced OPC dynamics (P180+) (28, 42). *Chst3*-deficient OPCs exhibited normal rates of attempted differentiation and integration at both ages, suggesting that this matrix change is not used in a cell autonomous manner to modulate their

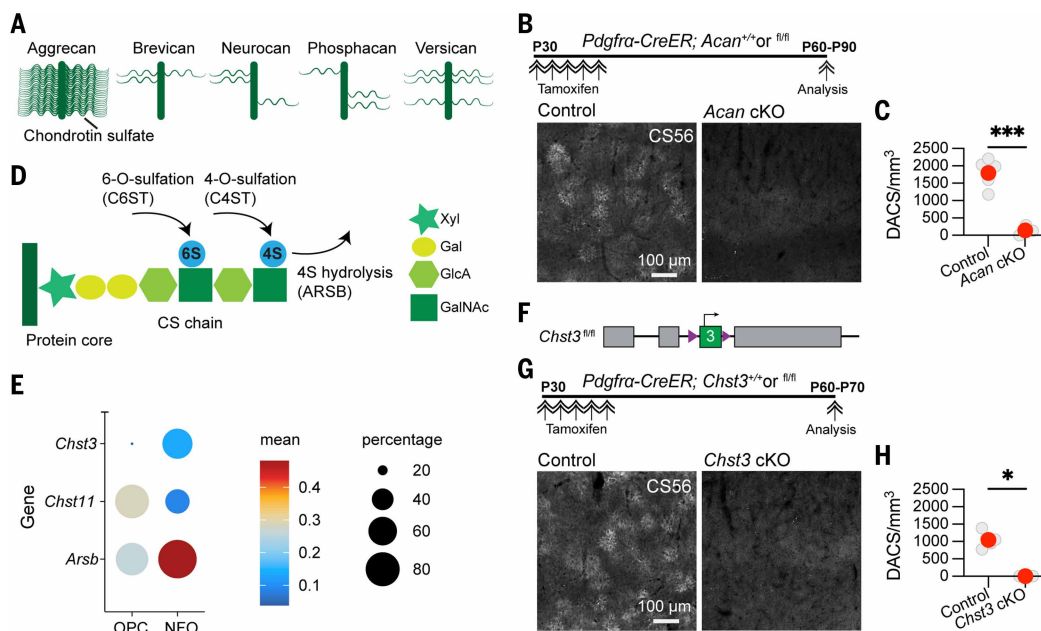


Fig. 3. DACS are 6-O-sulfated aggrecan secreted by OPCs. (A) Schematic representation of CSPGs and their CS chains. (B) Experimental strategy of conditional *Acan* ablation in the OL lineage (top). Tamoxifen injections were administered daily for 5 days at P30, and tissue was collected and analyzed at P60 to P90. Immunostaining of CS56 in the somatosensory cortex of the control and *Acan* cKO mice is shown at the bottom. (C) Quantification of CS56⁺ DACS in the somatosensory cortex of control and *Acan* cKO mice. Filled circles and error bars represent means $\pm$ SEM. Data are as follows: Control, 1798 $\pm$ 184 DACS/mm³, n = 5 mice; *Acan* cKO, 139 $\pm$ 84 DACS/mm³, n = 3 mice; p = 0.0003, unpaired, two-tailed Welch's t test. (D) Schematic representation of CS chain sulfation. CS chains are sulfated at the C-4 and/or C-6 position of GalNAc by sulfotransferases CS4T and CS6T, respectively. 4-O-sulfated CS is hydrolyzed by arylsulfatase ARSB. Gal, galactose; GlcA, glucuronic acid; Xyl, xylose. (E) Dot plot of *Chst11* (C4ST), *Chst3* (C6ST), and *Arsb* (ARSB) expression in quiescent and differentiating OPCs. The size of the circles indicates the percentage of the cells expressing the genes in each OPC subtype, and the color corresponds to the normalized mean expression amount. Data from the dataset are described in fig. S1A. (F) Schematic representation of a conditional *Chst3* allele. LoxP sites (purple triangles) flank exon 3, which contains a *Chst3* start codon (arrow). (G) Experimental strategy for conditional *Chst3* ablation in the OL lineage (top). Tamoxifen injections were administered daily for 5 days at P30, and tissue was collected and analyzed at P60 to P70. Immunostaining of CS56 in the somatosensory cortex of the control and *Chst3* cKO mice is shown at the bottom. (H) Quantification of CS56⁺ DACS in the somatosensory cortex of control and *Chst3* cKO mice. Filled circles and error bars represent means $\pm$ SEM. Data are as follows: Control, 1047 $\pm$ 100 DACS/mm³, n = 5 mice; *Chst3* cKO, 0 $\pm$ 0 DACS/mm³, n = 4 mice; p = 0.0159, Mann-Whitney test. *** p < 0.001; * p < 0.05.

differentiation (fig. S8, A and B). However, OPCs in *Chst3* cKO mice exhibited higher proliferation in both the young (P75+) and adult (P180+) brain, suggesting that this ECM modification may function in a cell-nonautonomous manner to control their homeostatic replacement, with each differentiating OPC signaling its removal from the population.

OPC differentiation attempts do not scale with oligodendrogenesis

The density of oligodendrocytes varies markedly across the brain, with defined gradients in oligodendrogenesis that persist across the life span (18, 19, 28, 35). By contrast, OPCs establish and maintain a relatively constant, tiled distribution across the CNS (20), and they are also present in regions where no oligodendrocytes or myelin are ever formed, such as the molecular layer of the cerebellum (72) (Fig. 4A). If differences in OPC lineage progression contribute to this myelin patterning, the density of DACS and oligodendrocytes should be correlated. However, DACS were present in the molecular layer of the cerebellum, suggesting that OPCs constantly attempt to differentiate in this area but never successfully integrate (Fig. 4B).

Although the density of DACS was lower in the cerebellum than in the somatosensory cortex, normalization to the lower density of OPCs in this area revealed that the rate of OPC differentiation was similar between these brain regions, suggesting cell-autonomous regulation of lineage progression (Fig. 4, C and D). In vivo two-photon imaging in *Pdgfra-CreER;RCE* mice confirmed that OPCs frequently attempt

to differentiate in the cerebellar molecular layer but ultimately die, with visible process fragmentation (fig. S9 and movies S4 and S5). Thus, OPC lineage progression is apparently uncoupled from oligodendrocyte need in the adult brain. This evidence of ongoing, attempted lineage progression is consistent with previous studies showing that increasing the diameter of granule cell axons in this region (73) or modifying the PUMA pathway in oligodendroglia (72), which controls apoptosis, results in the inappropriate appearance of oligodendrocytes in the molecular layer. Together, these results provide additional evidence for brainwide cell constitutive control of OPC lineage progression.

OPC differentiation attempts decline with age

Oligodendrogenesis begins after birth in the cerebral cortex and accelerates over the next several months before gradually declining in adulthood (28, 74, 75). To assess whether changes in OPC differentiation behavior contribute to this aging-dependent decline in oligodendrocyte generation, we compared DACS in the somatosensory cortex of P100 and P365 mice. DACS density was lower in older mice, despite a similar number of OPCs (Fig. 5, A and B). Consistent with this evidence of reduced lineage progression, there was a significant decrease in the number of DACS⁺ BCAS1⁺ cells, suggesting that OPC differentiation attempts decline with age (Fig. 5, A and B). To further explore aging-dependent changes in OPC lineage progression, we performed time-lapse in vivo two-photon imaging of OPCs in *Pdgfra-CreER;RCE* mice beginning at P75 and P180 (Fig. 5C). Analysis of these longitudinal datasets revealed

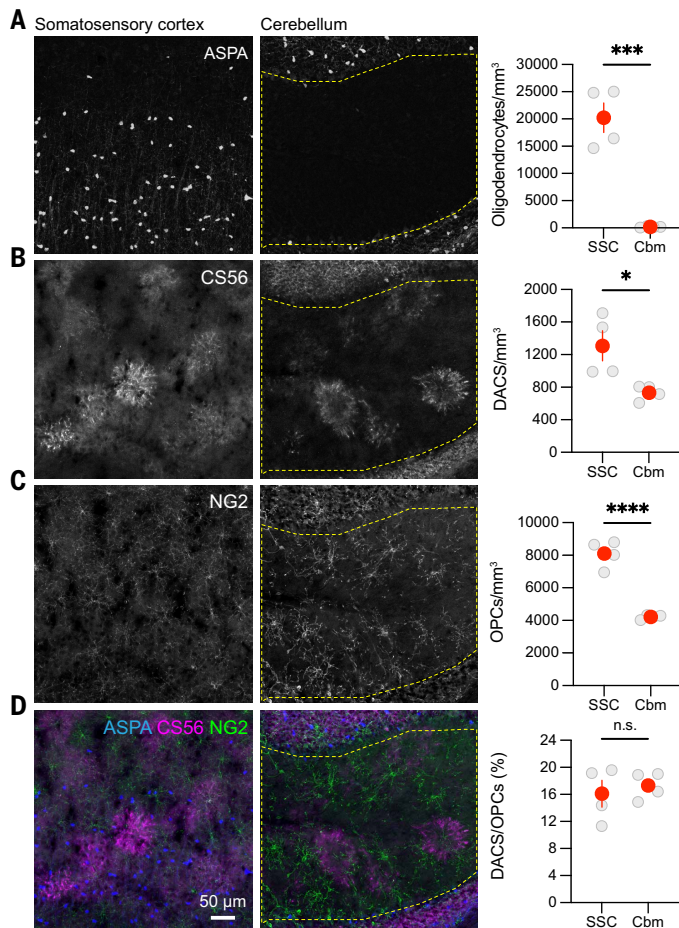


Fig. 4. OPC differentiation attempts are agnostic to myelination demands.

(A) Immunostaining of ASPA⁺ oligodendrocytes in the somatosensory cortex (SSC) and cerebellar molecular layer (Cbm; area within dashed lines) brain regions of a C57BL/6 13- to 15-week-old mouse. Quantification of ASPA⁺ oligodendrocyte density is shown at the right. Filled circles and error bars represent means $\pm$ SEM. Data are as follows: SSC, 20225 ± 275 cells/mm³, $n = 4$ mice; Cbm, 178 ± 52 cells/mm³, $n = 4$ mice; $p = 0.0003$, unpaired, two-tailed Student's t test. (B) Immunostaining of CS56 DCS in the somatosensory cortex and cerebellar molecular layer. Quantification of DCS density is shown on the right. Data are as follows: SSC, 1308 ± 185 DCS/mm³, $n = 4$ mice; Cbm, 732 ± 47 DCS/mm³, $n = 4$; $p = 0.0286$, Mann-Whitney test. (C) Immunostaining of NG2⁺ OPCs in the somatosensory cortex and cerebellar molecular layer. Quantification of NG2⁺ OPC density is shown on the right. Data are as follows: SSC, 8108 ± 420 cells/mm³, $n = 4$ mice; Cbm, 4214 ± 68 cells/mm³, $n = 4$ mice; $p < 0.0001$, unpaired, two-tailed Student's t test. (D) Merged images from (A) to (C). Quantification of DCS/OPC ratio is shown on the right. Data are as follows: SSC, $16 \pm 2\%$, $n = 4$ mice; Cbm, $17 \pm 1\%$, $n = 4$ mice; $p = 0.6096$, unpaired, two-tailed Student's t test. **** $p < 0.0001$; *** $p < 0.001$; * $p < 0.05$; n.s., not significant.

that OPCs in older mice showed reduced differentiation attempts, and ultimately, fewer oligodendrocytes were formed (Fig. 5, D and E). These results indicate that OPCs shift to a more quiescent state as the brain ages, consistent with the progressive decline in transitioning OPCs in the cortex (42, 76) and histological evidence of reduced homeostatic proliferation of OPCs across the life span (16).

OPCs do not increase differentiation in response to oligodendrocyte degeneration and demyelination

The mammalian brain exhibits a notable ability to recover from loss of oligodendrocytes through injury or disease by producing new

oligodendrocytes to regenerate myelin sheaths (77). This enhanced oligodendrogenesis has been assumed to arise from the rapid mobilization of OPCs to proliferate and differentiate in response to demyelination (78). If OPCs detect changes in oligodendrocyte viability or the presence of myelin, the frequency with which they differentiate should increase upon demyelination. To assess OPC behavior after oligodendrocyte death, we monitored OPC differentiation attempts using both DCS immunostaining and in vivo imaging in mice that had been fed the oligotoxin cuprizone (Fig. 6A). Tissue examined 2 weeks after cessation of cuprizone administration, a time when the demand for new oligodendrocytes and myelin is high, exhibited the same density of DCS as controls (Fig. 6, B and C). At this stage of recovery, there is a pronounced increase in oligodendrogenesis (77). Analysis of NFOs coimmunoreactive for CS56 and BCAS1 revealed that more of these cells exhibited morphology of newly integrated oligodendrocytes with nascent myelin sheaths (77), suggesting that DCS are increasingly associated with surviving NFOs during recovery (fig. S10). These results suggest that the apparent increase in oligodendrogenesis is due to enhanced survival of transitioning cells, rather than enhanced OPC lineage progression.

To provide an independent assessment of this lack of apparent OPC mobilization, we performed longitudinal in vivo two-photon imaging of OPC lineage progression in *Pdgfra-CreER;RCE* mice subjected to the same demyelination protocol (Fig. 6D). Consistent with the DCS analysis, the frequency of OPC differentiation attempts was not increased during the time when the appearance of new oligodendrocytes was maximal, and OPC differentiation remained consistent across cortical depth, despite the increase in appearance of new oligodendrocytes in deeper layers of the cortex (Fig. 6, E to G). The behavior of OPCs was similarly unchanged when oligodendrocytes were selectively ablated using DT (*Mobp-iCreER;Mobp-EGFP;Rosa26-lsl-iDTR* mice) (77, 79, 80) (fig. S11A). In this model, widespread loss of oligodendrocytes was observed in the somatosensory cortex 8 weeks after DT injection, yet the density of DCS remained similar to that of control mice, indicating that the lack of OPC mobilization is not due to unanticipated effects of cuprizone on these cells (fig. S11, B and C). Together, these results indicate that OPCs do not sense the presence (or loss) of oligodendrocytes and do not alter their rate of differentiation to promote myelin repair. Therefore, constitutive OPC lineage progression (34) continues to define their behavior after demyelination.

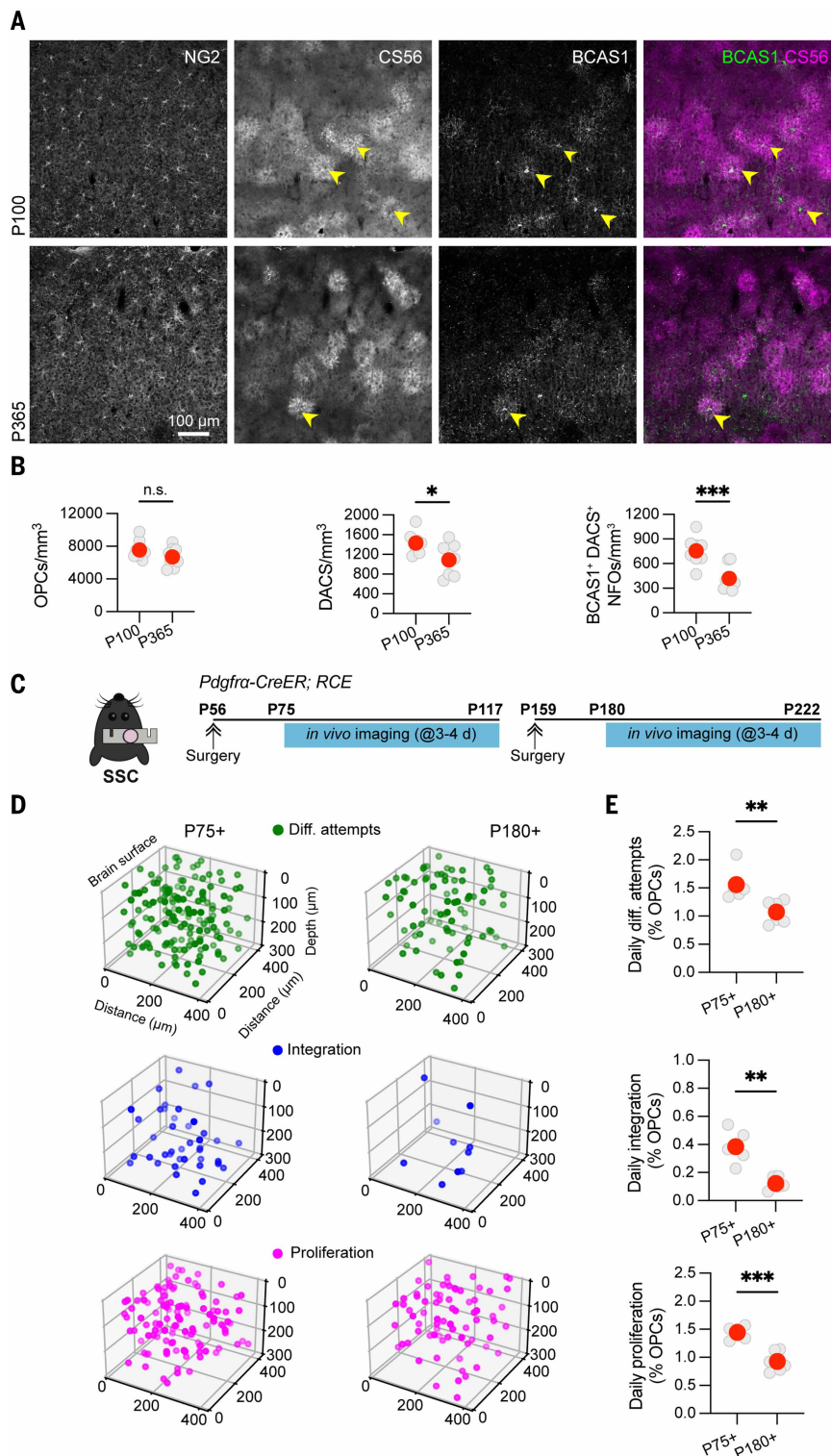
Inflammation reduces OPC differentiation attempts

The proliferation and differentiation of OPCs is affected by inflammatory cytokines, which can promote oligodendrogenesis in the early phases of remyelination but impair lineage progression in more severe chronic inflammatory states (81–83). To determine whether the cell-intrinsic regulation of OPC differentiation is affected by inflammation, we assessed differentiation attempts of OPCs in mice that had been exposed to a high dose of lipopolysaccharide (LPS), which induces rapid activation of microglia in the brain and a profound increase in the production of inflammatory cytokines (84) (Fig. 7A). Wild-type mice exposed to LPS progressively lost weight, their core temperature declined (85), and microglia in the cortex of these mice exhibited hypertrophy, indicative of an activated state (86) (Fig. 7, B to D). Although the density of OPCs was not affected by LPS, there were fewer DCS, indicating that persistent inflammation suppresses OPC differentiation attempts, which may limit their ability to replace oligodendrocytes in the context of injury and disease (Fig. 7, E and F).

OPCs prioritize homeostatic self-replacement over lineage progression

OPCs in the uninjured CNS follow two fates, either differentiating or proliferating in response to the attempted differentiation of another OPC to maintain a consistent progenitor pool. OPC depletion has been observed in active and chronic MS lesions (8, 87), but the effect of this

Fig. 5. Oligodendrocyte lineage progression slows with aging. (A) Immunostaining of NG2⁺ OPCs, BCAS1⁺ NFOs, and CS56 DACS in the somatosensory cortex of P90 to P120 and P365 to 375 C57BL/6 mice. Yellow arrowheads indicate DACS colocalizing with BCAS1⁺ NFOs. (B) Quantification of cell densities. For OPC density, data are as follows: P100, 7560 ± 402 cells/mm³, $n = 8$ mice; P365, 6705 ± 431 cells/mm³, $n = 8$ mice; $p = 0.2786$, Mann-Whitney test. For DACS density, data are as follows: P100, 1431 ± 77 DACS/mm³, $n = 8$ mice; P365, 1086 ± 114 DACS/mm³, $n = 8$ mice; $p = 0.0379$, Mann-Whitney test. For BCAS1⁺ DACS⁺ NFO density, data are as follows: P100, 754 ± 60 cells/mm³, $n = 8$ mice; P365, 419 ± 54 cells/mm³, $n = 8$ mice; $p = 0.0006$, Mann-Whitney test. (C) Experimental strategy for in vivo two-photon imaging in P75 and P180 *Pdgfra-CreER*; *RCE* mice. In both groups, tamoxifen was administered daily for five consecutive days at P30 to P35. Cranial windows were installed at P62 to P65 or P159, and every 3- to 4-day time-lapse imaging was performed 2 to 3 weeks after surgery. (D) A 3D plot representing locations of OPC cell bodies imaged in mouse somatosensory cortical volumes over 42 days. The cortical volume dimensions are as follows: $412 \mu\text{m}$ by $416 \mu\text{m}$ by $300 \mu\text{m}$ for P75+ and $405 \mu\text{m}$ by $419 \mu\text{m}$ by $300 \mu\text{m}$ for P180+. Each graph represents the total number of differentiation attempts, successful oligodendrocyte integrations, and proliferating OPCs per representative region. The $0\text{-}\mu\text{m}$ depth is brain surface. (E) Quantification of OPC differentiation attempts, oligodendrocyte integration, and proliferation daily rates in the top $300 \mu\text{m}$ of primary somatosensory cortex over 42 days of in vivo two-photon imaging. All values are normalized to the total number of OPCs in the imaging volume. Filled circles and error bars represent means $\pm$ SEM. For daily differentiation attempts, data are as follows: P75+, $1.6 \pm 0.1\%$, $n = 5$ mice; P180+, $1.1 \pm 0.1\%$, $n = 6$ mice; $p = 0.0043$, Mann-Whitney test. For daily integration, data are as follows: P75+, $0.4 \pm 0.1\%$, $n = 5$ mice; P180+, $0.1 \pm 0.0\%$, $n = 6$ mice; $p = 0.0063$, unpaired, two-tailed Welch's *t* test. For daily proliferation, data are as follows: P75+, $1.4 \pm 0.1\%$, $n = 5$ mice; P180+, $0.9 \pm 0.1\%$, $n = 6$ mice; $p = 0.0003$, unpaired, two-tailed Welch's *t* test. *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; n.s., not significant.



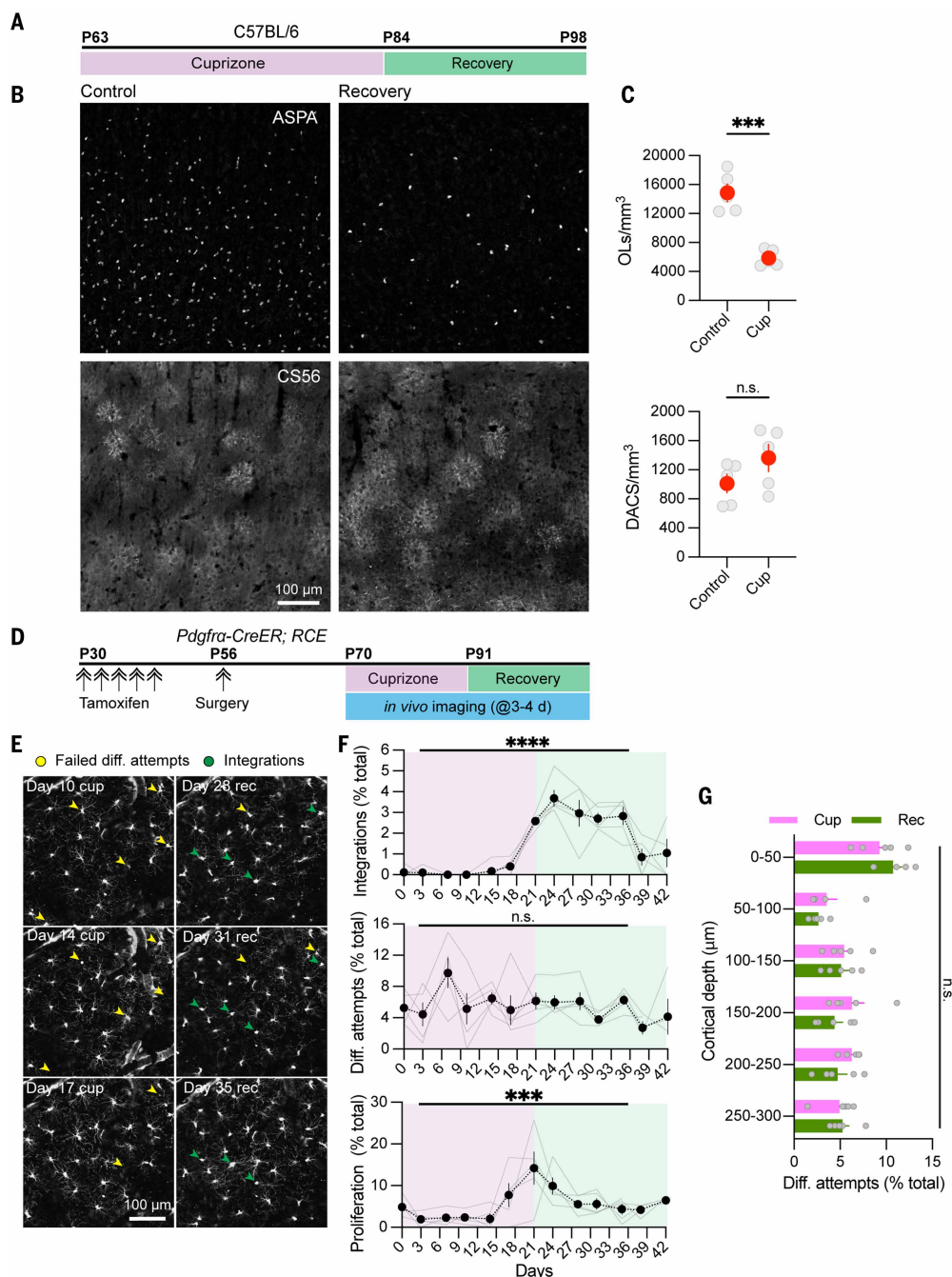
increased homeostatic pressure on their cell-intrinsic differentiation is not known. To assess how engaging in homeostatic proliferation affects their ability to differentiate, we examined the frequency of differentiation attempts in OPC-Terminator mice (*Olig2-Cre*; *OPC-Terminator*^{fl/+}) 15 to 17 days after DT administration, a time when prior OPC proliferation was high but OPC repopulation was now nearly complete (Fig. 7, G to I). Both DACS and BCAS1⁺ NFO density remained substantially reduced at this stage of recovery, suggesting that lineage progression is suppressed by homeostatic proliferation (Fig. 7, H and I).

Thus, direct damage to OPCs during chronic inflammation may further impair their ability to generate new oligodendrocytes, as they prioritize homeostatic replacement over differentiation.

Discussion

The persistence of OPCs in the adult CNS enables a prolonged period of developmental oligodendrogenesis, delaying circuit maturation until after adolescence and allowing additional myelin to be added to existing circuits in response to changes in life experience, a phenomenon

Fig. 6. OPC differentiation attempts do not increase after myelin loss. (A) Experimental strategy for inducing cuprizone-mediated demyelination and recovery in C57BL/6 mice brain. The pink bar indicates the duration of cuprizone administration with food, and the green bar indicates the time animals spent on the regular diet before tissue collection and analysis. **(B)** Immunostaining of ASPA⁺ oligodendrocytes and CS56⁺ DACS in control and cuprizone-fed C57BL/6 mice after 2 weeks of recovery. **(C)** Quantification of cell densities in the somatosensory cortex. Filled circles and error bars represent means $\pm$ SEM. For oligodendrocyte density, data are as follows: Control, 14858 ± 1217 cells/mm³, $n = 5$ mice; cuprizone, 5863 ± 506 cells/mm³, $n = 5$ mice; $p = 0.0001$, unpaired, two-tailed Student's t test. For DACS density, data are as follows: Control, 1011 ± 128 DACS/mm³, $n = 5$ mice; cuprizone, 1363 ± 186 DACS/mm³, $n = 5$ mice; $p = 0.1570$, unpaired, two-tailed Student's t test. **(D)** Experimental strategy for in vivo two-photon imaging during cuprizone and recovery in *Pdgfra-CreER;RCE* mice. Tamoxifen was administered at P30 before cranial window surgeries, and animals were imaged while undergoing cuprizone treatment and recovery. **(E)** Maximum intensity projection images (410 μ m by 407 μ m by 15 μ m) of the same region from a *Pdgfra-CreER;RCE* mouse somatosensory cortex imaged repeatedly during cuprizone and recovery. Yellow arrowheads indicate failed differentiation attempts, and green arrowheads indicate successful OL integrations. **(F)** Quantification of OPC integrations and differentiation attempts across 35 days in the top 300 μ m of the primary somatosensory cortex. Comparisons are made between days 3 to 21 (cuprizone) and days 24 to 35 (peak recovery). For each time point, values for each category of cells were normalized to the total number of tracked OPCs in a cortical volume. Filled circles and error bars represent means $\pm$ SEM, and thin lines represent values from individual mice. For all analyses, $n = 5$ mice, and the statistical tests used were the Friedman test with Dunn's post hoc test (integrations and differentiation attempts) and repeated measures one-way analysis of variance (ANOVA) with Tukey's post hoc test (proliferations). Statistical data are as follows: for integrations, Friedman statistic = 37.51, $p < 0.0001$; for differentiation attempts, Friedman statistic = 14.80, $p = 0.0965$; for proliferations, $F = 4.910$, $p = 0.0003$. Individual comparisons are listed in tables S4 to S6. **(G)** Quantification of the percentage of OPCs undergoing differentiation attempts across cortical depth in the somatosensory cortex during 21 days of cuprizone and 11 days of recovery. On the plot, 0 μ m denotes the top of the cortex. $n = 5$ mice. Data were analyzed by multiple Wilcoxon tests; q values (by depth, μ m), 0–50: 0.568125; 50–100: 0.984750; 100–150: 0.984750; 150–200: 0.631250; 200–250: 0.568125; 250–300: >0.999999. **** $p < 0.0001$; *** $p < 0.001$; n.s., not significant.



termed adaptive myelination (88, 89). Oligodendrogenesis is also enhanced by demyelination, allowing regeneration of oligodendrocytes and myelin destroyed through injury or disease (9, 90). Both adaptive myelination and myelin regeneration exhibit exquisite sensitivity, with myelin increasing specifically on the axons of active neurons (91) and new cells appearing in response to the loss of a single oligodendrocyte (92). The mobilization of OPCs necessary for this oligodendrogenesis has been assessed through changes in proliferation, through genetic fate tracing, by monitoring the up-regulation of genes specific to premyelinating NFO

stages (*Bcas1*, *Enpp6*) (61, 93, 94), and through the de novo appearance of oligodendrocytes and myelin internodes (21, 95). These results suggest that OPCs sense the presence of oligodendrocytes and the activity of surrounding neurons, and can be mobilized to generate additional oligodendrocytes and myelin. Indeed, OPCs form direct synapses with neurons, serving as postsynaptic partners that mark future sites of myelination (96, 97). However, our studies indicate that the initial steps of this process, which involve changes in gene expression associated with lineage progression and initial morphological transformation,

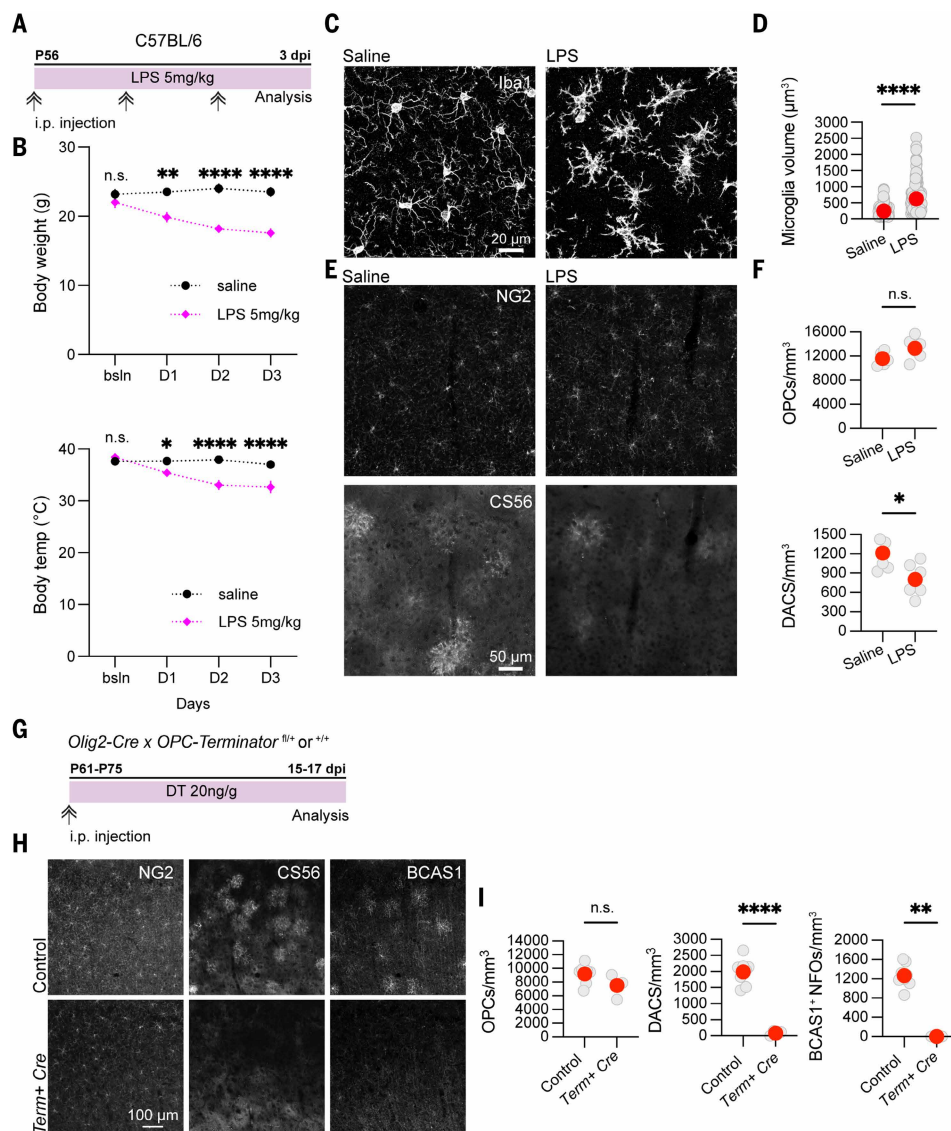


Fig. 7. OPC differentiation attempts are suppressed by inflammation and during homeostatic replacement.

(A) Experimental strategy of LPS injection in C57BL/6 mice. During the experiment, 5 mg/kg body weight LPS was administered to mice for three consecutive days. Control mice received saline injections. Tissue was collected and analyzed 20 hours after the third injection. dpi, days postinjection; i.p., intraperitoneal. (B) Quantification of body weight and core temperature in saline- and LPS-injected mice, $n = 6$ mice per group. For both quantifications, two-way repeated measures ANOVA with Sidak multiple comparisons was used. Data for body weight are as follows: $F = 80.82$, $p < 0.0001$; baseline (bsln) adjusted $p = 0.6843$, D1 adjusted $p = 0.0029$, D2 and D3 adjusted $p < 0.0001$. Data for body temperature are as follows: $F = 13.16$, $p < 0.0001$; bsln adjusted $p = 0.8489$, D1 adjusted $p = 0.0458$, D2 and D3 adjusted $p < 0.0001$. (C) Immunostaining of Iba1⁺ microglia in the somatosensory cortex. (D) Quantification of microglial cell body volume: saline, $246 \pm 8 \mu\text{m}^3$, $n = 251$ cells per six mice; LPS, $627 \pm 25 \mu\text{m}^3$, $n = 241$ cells per six mice; $p < 0.0001$, Mann-Whitney test. Analysis of data averages per animal is as follows: saline, $248 \pm 18 \mu\text{m}^3$; LPS, $625 \pm 42 \mu\text{m}^3$; $n = 6$ mice; $p = 0.0022$, Mann-Whitney test. (E) Immunostaining for NG2⁺ OPCs and CS56 in the somatosensory cortex. (F) Quantification of cell densities. For OPCs density, data are as follows: saline, 11562 ± 415 cells/mm³, $n = 6$ mice; LPS, 13259 ± 733 cells/mm³; $p = 0.0716$, unpaired, two-tailed Student's t test. For DACS density, data are as follows: saline, 1213 ± 104 DACS/mm³; LPS, 801 ± 105 DACS/mm³; $p = 0.0189$, unpaired, two-tailed Student's t test. (G) Experimental strategy of DT injection in *Olig2-Cre;OPC-Terminator* mice. One injection of 20 ng/g body weight DT was administered to mice. (H) Immunostaining of NG2, CS56, and BCAS1 in the motor cortex of control and *Olig2-Cre;OPC-Terminator^{fl/+}* mice ("Term + Cre") at 15 to 17 dpi. (I) Quantification of OPCs, DACS, and BCAS1⁺ cells densities in control and *Term + Cre* mice at 15 to 17 dpi. Filled circles and error bars represent means $\pm$ SEM. For OPC density, data are as follows: Control, 9196 ± 473 cells/mm³, $n = 8$ mice; *Term + Cre* DT, 7518 ± 754 cells/mm³, $n = 4$ mice; $p = 0.0727$, Mann-Whitney test. For DACS density, data are as follows: Control, 1987 ± 140 DACS/mm³, $n = 8$ mice; *Term + Cre* DT, 81 ± 33 DACS/mm³, $n = 4$ mice; $p < 0.0001$, unpaired, two-tailed Welch's t test. For BCAS1⁺ cell density, data are as follows: Control, 1263 ± 85 cells/mm³, $n = 8$ mice; *Term + Cre*, 0 ± 0 cells/mm³, $n = 4$ mice; $p = 0.0040$, Mann-Whitney test. **** $p < 0.0001$; ** $p < 0.01$; * $p < 0.05$; n.s., not significant.

occur consistently throughout the adult CNS, independent of the demand for additional oligodendrocytes and myelin. These results suggest that enhanced oligodendrogenesis—the appearance of new myelinating oligodendrocytes—is primarily achieved by enabling NFO survival (72, 73), rather than by increasing OPC differentiation. The nature of the signals that enable NFOs to exit this vulnerable stage and achieve full maturation are unknown.

The concept that developmental oligodendrogenesis is determined by the constitutive differentiation of OPCs was first proposed by Raff and colleagues, who showed that OPCs (O-2A progenitors) acutely isolated from the embryonic optic nerve were capable of spontaneous differentiation after a fixed number of divisions in vitro (33). This analysis was extended in vivo by Richardson and colleagues, who demonstrated that there is widespread proliferation of OPCs and efficient integration of NFOs in the developing spinal cord (34, 37). However, it has been unclear whether this constitutive program persists in the adult CNS or is replaced by regulated recruitment. Cumulative tracking of dividing OPCs with thymidine analogs [5-bromo-2'-deoxyuridine (BrdU), EdU] in the adult brain revealed widespread, linear proliferation of the population, consistent with persistence of this developmental program. However, direct visualization of OPC dynamics in the adult brain through long term in vivo two-photon imaging revealed that OPC proliferation varies considerably among different cells, between brain regions, and with the age of the animal (Fig. 5, D and E). Further, use of homeostatic proliferation as a proxy for lineage progression is complicated by the turnover of OPCs (57), direct differentiation of OPCs without proliferation (15, 18, 57, 98), and the potential toxicity of thymidine analogs used for tracking OPC proliferation (16, 18, 19, 34, 37, 99). Our studies show that increased OPC proliferation after ablation is associated with lower, rather than higher, lineage progression, suggesting that increased OPC proliferation is not always a reliable predictor of differentiation in vivo (Fig. 7 and fig. S4). In keeping with the PDGF-AA "supply-and-demand" model (37), which proposes that the dynamics of proliferation and lineage progression are determined by the availability of PDGF-AA, OPCs may prioritize homeostatic proliferation over lineage progression in these conditions until PDGF-AA amounts drop as the cells reach their preferred density.

We found that OPCs orchestrate a distinct alteration of the surrounding ECM as they begin to differentiate, acutely altering the

sulfation pattern (from the C-4 to C-6 position) on aggrecan before ceasing expression of this matrix molecule by the time they form myelin sheaths. The position of negatively charged sulfur atoms on CSPGs has been shown to create a “sulfation code” that influences their interaction with growth factors and cell-surface proteins to control cell growth (100). C-6-sulfation predominates in the developing CNS (100) and promotes axon regeneration when reestablished in the adult CNS (101). Changes in CSPG sulfation have been shown to alter the binding and dispersion of growth factors and protein-CSPG interactions (102–104). By acutely altering the surrounding ECM, a phenomenon conserved from mouse to human, OPCs may alert nearby OPCs that differentiation has occurred to promote temporal and spatial pacing of oligodendrogenesis. This alteration of signal dispersion by differentiating OPCs may be responsible for the changes in homeostatic proliferation observed when CSPG sulfation was selectively disrupted.

Although in vivo imaging provides a direct measure of lineage progression, the high technological cost and expertise required for these studies, and the inaccessibility of many brain regions to this procedure, limit its broad application. This new insight into the modification of the ECM by transitioning OPCs not only reveals their steady differentiation across the CNS but also provides a potential approach to assess the initial stages of oligodendrogenesis in physiological and pathological contexts. The continuous differentiation of these progenitors (and their homeostatic replacement) suggests that OPCs are among the most numerous cells produced in the CNS across the life span. However, estimates of OPC differentiation using DACS require knowledge about their time course of formation and clearance. Induction and cessation of conditional ChABC expression revealed that DACS continuously turn over in the brain, but because matrix formation and modification can be affected by age, infection, and injury (105, 106), it will be important to consider these possible changes when inferring OPC differentiation rates solely from DACS densities.

Our results suggest that oligodendrocyte “regeneration” might be an extension of developmental oligodendrogenesis, rather than a distinct reparative process. Consistent with this conclusion, oligodendrocytes and myelin sheaths appear in new locations after demyelination (77), recapitulating developmental myelin patterning rather than restoring the original sheath distribution. Moreover, oligodendrocytes exhibit extraordinary stability and longevity (74, 75); in the absence of regular oligodendrocyte turnover or degeneration, there may have been insufficient evolutionary pressure to establish a separate regenerative process involving OPC mobilization, as demyelinating injuries and disease are rare in early life (107). The decline in frequency of OPC differentiation attempts with age, in response to severe inflammation, and after homeostatic replacement could contribute to the slower recovery observed in patients with demyelinating diseases such as MS (108). Nonetheless, DACS are present in the brain of individuals well into the eighth decade of life (44), indicating that in humans, OPC differentiation attempts continue throughout life. Leveraging this knowledge about initial transcriptional changes and ECM modification by OPCs provides a means to analyze OPC differentiation rates in the human CNS, which may accelerate our understanding of their behavior in MS and spur the development of new approaches to extend this developmental program for therapeutic benefit.

Materials and methods

Animal care and use

All animal experiments were conducted in accordance with the National Institute of Health Guide for the Care and Use of Laboratory Animals and were approved by Animal Care and Use Committee at Johns Hopkins University (Protocol numbers: MO23M202, MO20M308). Whenever possible, female and male mice of specified ages were used for experiments and randomly assigned to experimental groups. Mice were maintained in a climate-controlled room on a 12 hour light-dark cycle, housed in groups no larger than five, and food and water were

provided ad libitum unless animals were receiving cuprizone, EdU, and doxycycline treatments as described below. All mice were maintained on C57BL/6 (B6) background. Generation and genotyping of transgenic lines *Mobp-EGFP* (Gensat) (15), *Mobp-iCreER* (79), *Pdgfra-CreER* (17), *Olig2-Cre* (109), *Rosa26-lsl-EGFP* (RCE) (60), *Rosa26-lsl-mGamp6s* (59), *Acan^{fl/fl}* (EUCOMM) (66), *CAMKII-tTA* (49), and *Rosa26-lsl-iDTR* (80) have been previously described. WT C57BL/6 were purchased from Charles River. *Matn4-mEGFP* mouse line generation and genotyping were described in detail elsewhere (42). Heterozygous animals were used for *Cre* and reporter lines.

Generation of transgenic mouse lines

Transgenic mouse lines were generated by the Johns Hopkins transgenic facility core. *ChaseR-TRE* donor plasmid was generated by subcloning a coding sequence of mammalian optimized ChABC (48) into a pTRE-Tight cloning vector (Addgene no. 50547) (110), a 3xHA tag sequence in frame with ChABC, and WPRE. The plasmid was linearized, and the resulting donor sequence was injected into one-cell C57BL/6J embryos to generate a transgenic line (111). Founders were screened by genotyping via polymerase chain reaction (PCR) using genomic DNA isolated from tail snips. *Xylt1* primer products were used as an internal positive control resulting in a 250 bp product and *ChaseR*⁺ specific product is 450 bp. The following genotyping primers were used: *ChaseR* F 5' TGCTACCCCTGTACAAATCA 3'; *ChaseR* R 5' GGCATTAAAGCAGCGTATCC'. *Xylt1* F 5' AGCATTTCTCCAGCCAAAGA 3'; *Xylt1* R 5' CTGGGACTCCATGACCTGTT 3'. The following PCR protocol was used: 95°C for 3 min, (95°C for 10 s, 58°C for 10 s, 72°C for 10 s) times 35 cycles, 72°C for 5 min using the KAPA Express PCR kit in a PCR thermocycler (Bio-Rad). Animals with positive genotyping results were backcrossed once onto a C57BL/6 background and F₁ progeny were screened for copy number using qPCR of genomic DNA. Selected founders were then crossed with *CAMKII-tTA* transgenic mice to determine the efficacy of ChABC expression. A mouse line (no. 446) that demonstrated the highest and most consistent ChABC expression as evidenced by immunolabeling for CSPG digestion and HA expression in forebrain neurons (fig. S2) was used for experiments in this study. *ChaseR-TRE* mice are viable and fertile and have a life span comparable to that of control littermates. For generation of a conditional *Chst3* allele, an 840 bp *Chst3* genomic sequence with exon 3 flanked by LoxP sites and 100 bp homology arms was assembled in silico and a single-stranded DNA was generated (IDT). Two guide RNAs (IDT) were designed in silico and transgenic animals were generated using Easi-CRISPR approach (112). Founders were screened for positive insertions with *Chst3^{fl/+}* specific primers, with a WT product of 281 bp and floxed product of 315 bp. Primer sequences were as follows: F 5' ACCACAGGTCCCTTTATCC 3'; R 5' CCCCCACATTAACTTTTCC 3'. PCR conditions were the same as listed above. Animals with positive insertions were confirmed by sequencing of PCR amplified *Chst3* genomic sequences and backcrossed onto a C57BL/6 background. F₁ progeny were validated for successful insertions with an additional round of PCR products sequencing. One line (no. 3581) was continued to be backcrossed onto BL/6 background and eventually bred to *Chst3^{fl/fl}* homozygosity. *Chst3^{fl/fl}* mice are viable and fertile and have a life span comparable to that of control littermates. For generation of an *OPC-Terminator* allele, a 4480-bp sequence was subcloned into a donor plasmid containing the following elements: (i) a splice acceptor site of the engrailed 2 (*En2*) gene; (ii) a T2A sequence; (iii) a *Terminator* cassette (consisting of a simian DTR sequence with a 3xHA tag, T2A sequence, thymidine kinase sequence with a 3xHA tag); (iv) two sets of incompatible LoxP sites flanking elements 1 to 3; (v) WPRE; (vi) polyA sequence, and (vii) 100 bp homology arms to the intron 1 of the *Cspg4* gene. Sequences of elements 1 to 3 are in reverse complement orientation to the *Cspg4* promoter. A linearized double-stranded DNA fragment of *OPC-Terminator* along with 2 guide RNAs (IDT) designed in silico were utilized in generation of transgenic animals with Easi-CRISPR approach (112). Founders were

screened for positive insertions with *OPC-Terminator* specific primers, with a WT product of 370 bp and floxed product of 200 bp. Primer sequences were as follows: Term LA F2 5' TGTCACCTTTGCTCCATCAGC 3'; Term RA R1 5' GTCTGAGAAGGCAGGGTGAG 3'; Term LA R1 5' ACGAAGTTATTCCGGAGTCG 3'. PCR conditions were the same as listed above. Animals with positive insertions were confirmed by sequencing of PCR amplified *Terminator* genomic sequences and backcrossed onto a C57BL/6 background. F₁ progeny were validated for successful insertions with an additional round of PCR products sequencing. One line (no. 513) continued to be backcrossed onto BL/6 background. In the absence of DT, *OPC-Terminator*^{fl/+} mice are viable and fertile and have a life span comparable to that control littermates.

OPC single-cell transcriptional profiling

OPC isolation, cDNA library preparation, and analysis are described in detail elsewhere (42).

Tamoxifen administration

Tamoxifen solution was prepared fresh before each experiment by dissolving tamoxifen powder (Sigma-Aldrich, T5648) in 100% sunflower seed oil (Sigma-Aldrich, S5007) at 10 mg/ml concentration, by sonication and vortexing the solution in 5 to 10 min intervals. For experiments requiring a saturating number of cells with gene recombination, four or five daily sequential injections of tamoxifen at 100 mg/kg body weight were given intraperitoneally to both control and experimental animals. For sparse recombination, animals were given a single 100 mg/kg body weight injection.

Doxycycline control of transgenic expression

Doxycycline hyclate (DOX, Sigma-Aldrich, D9891) was dissolved in distilled water at 50 mg/liter concentration and stored covered at RT for one week. Unless otherwise specified, animals were kept on DOX supplemented water in red-tinted bottles provided ad libitum from conception until animal weaning at P21 to P28. Water was replaced at least once a week. For ChABC ON/OFF experiments, dox supplemented water was reintroduced to both control and experimental animals for 14 days before tissue collection.

EdU administration

2'-Deoxy-5-ethynyluridine (EdU) was administered as described previously (54). EdU (Carbosynth, NE08701) was dissolved to a final concentration of 200 µg/ml with 1.5% sucrose in distilled water and provided to animals as drinking water in red-tinted bottles ad libitum as described previously (113). Solution was refreshed every 48 to 72 hours. To achieve maximum labeling of proliferating cells in *ChaseR-TRE*; *CAMKII-tTA*, additional intraperitoneal injections of EdU were administered twice a day at least 8 hours apart. EdU (ThermoFisher Scientific, E10187) was dissolved in 0.9% sterile saline (icumedical, NDC 0990-7983-03) and final administered concentration was 5 µl/g body weight.

Cuprizone administration

Animals were fed a diet of milled, irradiated 18% protein rodent diet (Teklad Global, 2918) supplemented with 0.2% w/w bis(cyclohexanone) oxaldihydrazone (cuprizone, Sigma-Aldrich, C9012) in custom gravity-fed food dispensers for 3 weeks. Cuprizone chow was made fresh every week and stored in airtight container with desiccator packets at 4°C when not in use. Feeders were refreshed 2 or 3 times a week. Control animals received chow only. Animals' weight was monitored for any signs of distress. During recovery, standard pellet rodent food was reintroduced.

Diphtheria toxin administration

Diphtheria toxin (Sigma-Aldrich, D0564) stock solution was prepared at 1 mg/ml concentration in sterile 0.9% saline. Stock solution was kept at -80°C and working solution was prepared by further dilution to 1 ng/µl concentration with 0.9% sterile saline. P110 to P124 male and female

Mobp-EGFP; *Mobp-iCreER*; *Rosa26-lsl-iDTR* mice in the experimental group received daily intraperitoneal injections for five consecutive days at 10 ng/g body weight. Control animals received 0.9% saline injections under identical conditions. *Olig2-Cre*; *OPC-Terminator*^{fl/+} mice received a single 20 ng/g body weight intraperitoneal injection of a 4 ng/g stock prepared as described above. Control group (with and without *Olig2-Cre*) included *OPC-Terminator*^{+/+} littermates that received DT injections and *OPC-Terminator*^{fl/+} littermates that received saline injections under identical conditions. Animals were monitored for weight loss and signs of distress and provided a DietGel (clear H20) for additional nutrition and hydration.

LPS administration

Lipopolysaccharide (Sigma-Aldrich, L6143) was dissolved in 0.9% sterile saline at 5 mg/kg LPS concentration and 8- to 9-week-old C57BL/6 mice (Charles River) were injected with either saline or 5 mg/kg LPS for three consecutive days, each injection at least 20 hours apart. Body temperature and body weight were measured immediately prior to each injection.

Cranial window surgery and in vivo two-photon imaging

Cranial window surgery: Cranial windows were prepared as previously described (77). Briefly, P45 to P56 (somatosensory cortex windows), P70 (cerebellar windows), P70 (somatosensory cortex windows, additional group), P159 (P180+ imaging group somatosensory cortex) mice were anesthetized with isoflurane (induction, 5%; maintenance, 1.5 to 2%, mixed with 0.5 liters/min O₂, VetOne, NDC 10019-360-60), and their body temperature was maintained at 37°C with a thermostat-controlled heating plate. Lidocaine was subcutaneously injected in the scalp (0.5%, VetOne, NDC 13985-222-04) to locally anesthetize. For the somatosensory cortex, the skin over the right hemisphere was removed and the skull cleaned. A 3 × 3 mm circular region was scored over the right side of the skull at coordinates -1.5 mm posterior and 3.5 mm lateral from bregma. For motor cortex, coordinates were 0.5 anterior and 2 mm lateral from bregma. A custom metal plate with a hole centered over a scored region was attached to the skull with dental cement (C&B Metabond, Parkell). A scored region of skull was removed using a high-speed dental drill. A piece of cover glass (Warner instruments, no. 1, CS-3R) was placed in the craniotomy, stabilized in place with a custom needle holder, and sealed with dental cement. For cerebellar windows, scalp and initial portion of the skin on top of the neck were excised, exposing the cervical part of the trapezoid muscle and the skull. The periosteum was removed, and trapezoids were detached from the skull, exposing most of the interparietal region. The splenic capitis muscle was cut to expose the rest of the interparietal region. Once a head bar was secured to the skull, a 2 × 2 mm region was scored at -6 anterior and at 0 medial/lateral from bregma. A 2 × 2 mm square custom-made coverslip was placed and the rest of the surgery proceeded as described above.

In vivo imaging: After 2 to 3 weeks of recovery, animals subjected to longitudinal in vivo two-photon imaging every 3 to 4 days or as otherwise indicated. Two-photon laser scanning microscopy was performed with a Zeiss LSM710 (Zeiss) microscope equipped with a GaAsP detector using a mode-locked Ti:Sapphire laser (Coherent Chameleon Ultra II) tuned to 920 nm. Mice were imaged under isoflurane anesthesia (same parameters as above), with head immobilized by attaching a head plate to a custom stage on the microscope table. One or two non-overlapping regions of interest (ROI) were selected based on window clarity. Fluorescence images were collected 0 to 350 µm from the cortical surface using a coverslip-corrected Zeiss 20x/1.0 W Plan-Apochromat objective. For improved registration of time-lapse imaging, *Matn4-mEGFP* mice were injected retro-orbitally with fluorescent dextran (20 µl, 2.5%, MW = 70 kDa), conjugated with either Texas Red or Rhodamine (Thermo Fisher Scientific D184) (114) to label the microvasculature. A 2.5% solution

was prepared by dissolving stock dextran crystals in sterile saline and kept in the dark at 4°C. Mice were injected under isoflurane anesthesia and were imaged 5 min after injection. For simultaneous acquisition of EGFP and dextran signals, the laser was tuned to 910 nm.

Tissue preparation, processing, immunohistochemistry, and imaging

Tissue preparation: Animals were deeply anesthetized with an intraperitoneal injection of pentobarbital (100 mg/kg body weight) and transcardially perfused with 10 to 15 ml 1x phosphate buffered saline, pH 7.4, followed by 25 ml of freshly prepared, ice-cold 4% paraformaldehyde/1x PBS (PFA, Electron Microscopy Sciences, 15714-S). Dissected tissues were postfixed in 4% PFA/PBS at 4°C for either 8 hours or overnight, and then cryoprotected in 30% sucrose/0.1% sodium azide in 1x PBS at 4°C for minimum of 48 hours. Brain tissue was embedded with Tissue-Tek O.C.T Compound (Sakura Finetek, 4583) and sectioned on Leica CM1950 cryostat (Leica) at 35 μ m thickness (sagittal sections in all experiments with the exception of coronal sections in Figs. 2, E and F, and 6, A to C) and stored as free-floating sections in 1x PBS/0.1% sodium azide.

Immunohistochemistry: Immunohistochemistry experiments were performed as described previously (79). Briefly, brain sections were washed at RT on a shaker for 3x10 min with 1xPBS, permeabilized in 0.3% Triton-X in 1xPBS for 10 min, then incubated in blocking solution containing 10% normal goat serum (NGS, Cell signaling, 5425S) + 0.3% Triton-X in 1x PBS for 1 hour at RT on a shaker. For ASPA staining, an additional 10-min incubation step at RT with no agitation in liberate antibody binding solution (L.A.B., Polysciences, Inc., 24310-500) was included after the first PBS wash. For primary antibody incubation, antibodies were diluted in blocking solution at the following concentrations: 1:1000 rabbit anti-ASPA (GeneTex, 113389, RRID:AB_2036283), 1:1500 mouse anti-CS56 (Sigma-Aldrich, C8035, RRID:AB_476879), 1:5000 guinea pig anti-NG2 (custom Bergles lab) (57), 1:1000 rabbit anti-BCAS1 (Synaptic Systems, 445 003, RRID:AB_2864793), 1:4000 chicken anti-GFP (Aves, GFP-1010, RRID:AB_2307313), 1:1000 rabbit anti-Iba1 (Wako, 019-19741, RRID:AB_839504), 1:1000 rabbit-anti HA (Cell Signaling, C29F4, RRID:AB_1549585), 1:1000 biotinylated WFA (Vector Biolabs, B-1355, RRID:AB_2336874). Sections were incubated on a shaker overnight at 4°C. The following day, sections were washed at RT on a shaker for 3x10 min in 1xPBS and secondary antibodies were applied at 1:1000 dilution in blocking solution for 2 hours on a shaker at RT in the dark. The following secondary antibodies and reagents were used in this study: 488, Cy3, and Cy5 donkey anti-guinea pig IgG (all Jackson ImmunoResearch; 488: 715-546-150, RRID:AB_2340849; Cy3: 715-166-151, RRID:AB_2340817; Cy5: 715-175-151, RRID:AB_2340820), Alexa Fluor 546 and Alexa 647 donkey anti-rabbit IgG (546: ThermoFisher Scientific, A10040, RRID:AB_2534016; 647: Jackson ImmunoResearch, 711-606-152, RRID:AB_2340625), 546 Alexa goat anti-mouse IgM (ThermoFisher Scientific, A21045, RRID:AB_1500928), 650 DyLight goat anti-mouse IgM (ThermoFisher Scientific, SA5-10153, RRID:AB_2556733), Alexa Fluor 488 donkey anti-chicken IgG (Jackson ImmunoResearch, 703-546-155, RRID:AB_2340376), and 555 Streptavidin (ThermoFisher Scientific, S32355, RRID:AB_2571525). Afterward, sections were then washed 3x10 min in 1x PBS, mounted on Superfrost coated slides (Fisher Scientific, 12-550-15), air dried, covered with Aqua-Poly/Mount (Polysciences, 18606-5) and coverslipped (VWR, No 1.5, 48393-251, 48404-466). For EdU detection, the Click-iT EdU Proliferation Kit for Imaging (ThermoFisher Scientific, C10340) was used according to manufacturer's instructions after secondary antibody incubation together with AlexaFluor 555 Azide (ThermoFisher Scientific, A20012). Final washes were extended to 6x10 min in 1x PBS to reduce staining background.

Tissue imaging: For cell density analysis, two or three nonoverlapping brain sections containing a region of interest were imaged using an inverted Zeiss confocal LSM800 with 10x/20x air objectives (Zeiss).

For control and experimental groups, special care was taken to acquire images using the same laser power and gain acquisition settings. FOV images were acquired as *xyz* tiles and stitched using Zeiss Zen Blue software (Zeiss, RRID:SCR_013672). For individual cell reconstructions and high magnification representative images, a 40x oil objective was used. Images of entire sagittal brain sections were acquired using an epifluorescence microscope Keyence BZ-X710 (Keyence) with a 20x air objective. Individual images were then stitched into a single file using Image Viewer software (Keyence).

Preparation of figures

Representative images were chosen from the same tissue sets used for analysis based on their representative value and quality. Brightness and contrast were adjusted for clarity using Fiji (ImageJ, NIH, RRID:SCR_002285) and special care was taken to ensure identical adjustment parameters for images of control and experimental groups. Confocal images were presented either as *z*-stack maximum intensity projections or optical sections at an optimal focal plane. All figures were generated in Adobe Illustrator 2024-2025 (Adobe, RRID:SCR_010279).

Histological data analysis

All data analysis was performed blinded to the sample condition.

Cell-density quantification: Confocal *z*-stacks of two or three nonoverlapping brain sections containing a region of interest were analyzed using Fiji and Imaris (Bitplane, RRID:SCR_007370), and an average number used as a single biological replicate, unless otherwise specified. Quantification was restricted to the top 600 μ m of cerebral cortex. For analysis of cell density in the cerebellar molecular layer, the entire molecular layer of a sagittal section was included in the analysis.

DACS analysis: As previously described (44, 45), DACS are round clusters of CS56 signal of approximately 70 to 100 μ m in diameter. For quantification, only DACS at least 50 μ m of diameter were quantified to avoid including out-of-plane DACS. Previously reported classification of DACS (or CS56 clusters) (45) included rosette and diffuse CS56 signal clustering depending on their appearance. For purposes of this study, all classified subtypes were included in quantification as DACS without distinction.

DACS/EGFP⁺ OL lineage colocalization: Colocalization analysis of the OL lineage cells and DACS was performed using Imaris. CS56 signal was used to generate a DACS surface volume and EGFP soma signal for cell surface volume. Only DACS and EGFP centroids whose coordinates were within 5 μ m of each other in *x*, *y*, and *z* were classified as double-positive.

OPC reconstruction and analysis: High resolution *z*-stacks of individually imaged EGFP⁺ cells were analyzed using Imaris. Representative reconstructions of individual EGFP⁺ OPCs were performed using Filament Tracer. Surface volumes of somas of analyzed cells were generated to measure the mean fluorescent intensity of somatic NG2 immunoreactivity. To account for inherent variability of histological staining, the mean fluorescence intensity of NG2 immunoreactivity of the soma in cells of interest was normalized to the same value measured in the nearest NG2⁺ DACS[−] OPCs. The number of primary branch points was determined by manual quantification in Imaris.

Nearest neighbor analysis of DACS and proliferating OPCs: Confocal 20x *z*-stack images were converted to a maximum intensity projection using Fiji. Analysis was performed using adjacent serial 35 μ m sections. XY coordinates of manually identified NG2⁺ OPCs, EdU⁺ cells, and DACS were determined. Adjacent sections were aligned using anatomical landmarks and DACS positions. We identified the closest six OPCs and the closest EdU⁺ OPC to each DACS in each section. For

quantification, if a DACS had at least one EdU⁺ OPC among its six nearest OPCs across either section, it was identified as having an EdU⁺ nearest neighbor (NN). As a control, we generated a random spatial distribution of DACS across the serial sections, maintaining the same distribution of OPCs and EdU⁺ OPCs. The centroids of these random DACS positions were generated using an unseeded uniformly distributed random number generator to select *x*- and *y*-coordinates for their positions. The number of random centroids was matched to the number of DACS observed in each respective section. NN calculations then proceeded the same as with actual DACS. Absolute distance measurements between DACS centroids and nearest EdU⁺ OPCs were calculated. The same analysis was performed on randomly distributed centroids. All analysis was performed using a custom-built MATLAB code (MATLAB, MathWorks, RRID:SCR_001622).

Microglia morphology analysis: Imaris software was used to quantify microglia cell body size. An ROI of size 200 × 200 μm was randomly picked for each section. In this ROI, the “surfaces” tool in Imaris was used to generate volume renderings of microglia visualized by positive staining of Iba1.

In vivo time-lapse imaging analysis

Z-stacks imaged across multiple time points were aligned by time, concatenated, and registered into a *xyz*t hyperstack using a Correct 3D drift plugin in Fiji (ImageJ, NIH). Registered hyperstacks were trimmed to the final volume ensuring that every frame contains visible data. For images containing labeled vasculature, additional registration to the channel containing vasculature imaging was performed with ITKElastix in Python (Python Software Foundation) (115). Trimmed hyperstacks were converted into h5/XML format using Big Data viewer plugin in Fiji. Fates of hundreds of individual cells were tracked in *xyz*t hyperstacks using a Fiji plugin MaMuT (116). Cell classification was performed as follows: (i) a cell was designated as “proliferating” at a time point following which two daughter cells were observed in its place; (ii) a cell was designated as “dead” at a time point following which it could no longer be found in the imaged volume and/or EGFP⁺ debris was observed; (iii) a cell was designated as “integration” at a time point when it exhibited substantial morphological transformation as described previously (57), surrounding OPCs were invading its territory, and it could be observed as smaller stationary cell at all following time points; (iv) “differentiation attempt” was calculated as a sum of (ii) and (iii). Generated files containing information in individually tracked cells were then analyzed with a custom analysis pipeline (Python). Z dimension was corrected for brain surface curvature. To account for different numbers of EGFP⁺ cells induced through Cre-mediated RCE recombination, the number of proliferations, differentiation attempts, and integrations at any given time point were normalized to the total number of imaged OPCs in that tissue volume at the corresponding time point. In *z* dimension, analysis was restricted to the top 300 μm in *Pdgfra-CreER;RCE* mice in the somatosensory cortex, to the top 100 μm in the cerebellar molecular layer, and to the top 350 μm in *Olig2-Cre;Matn4-mEGFP;OPC-Terminator^{fl/+}* mice. Integrated OLs were not counted toward total cell number after their initial designation. Cells exiting and entering an imaging FOV were included in the total cell number. For longitudinal analysis between experimental groups, data were aligned to a standardized timeline of alternating 3 or 4 days. Data from animal cohorts in experiments described in Fig. 5, C to E, were used as respective age-matched controls for experiments described in Fig. S8.

OPC analysis by depth: For analysis of OPCs across cortical depth during cuprizone and recovery, total numbers of differentiation attempts from time points 0 to 21 (cuprizone) and >21 to 35 (recovery) were pooled, parsed according to their *z*-depth in the imaged volume and normalized

to the total number of OPCs analyzed at those cortical depths and time points.

Analysis of OPC differentiation attempts over 10 days: Total number of OPC differentiation attempts over 10 days in imaged volumes from days 28 to 38 of imaging were summed and normalized to the total number of OPCs in the volume present on the 10th day. The number of DACS were quantified in top 375 μm of somatosensory cortex from two sagittal sections of *Pdgfra-CreER;RCE* mice and normalized to the number of NG2⁺ cells in those volumes.

Statistical analysis

No statistical tests were used to predetermine sample sizes, but our sample sizes are similar to those reported in previous publications (57, 79). Statistical analyses were performed using GraphPad Prism 10 (GraphPad). Shapiro-Wilk normality test was performed on data prior to comparing variables. For all datasets, the number of biological replicates, statistical tests performed, and their relevant values are listed in figure captions and tables S1 to S6, as indicated.

Sequencing analysis

Mouse visual cortex (V1) single-cell RNA sequencing data were downloaded from GEO (GSE102827), reanalyzed using Monocle 3 and visualized using Seurat (v3.2.2) as detailed in (38). Preannotated cell clusters in (38) were regrouped into following cell groups: OPC (OPC_1, OPC_2), NFO (Olig_6, Olig_7), OL (Olig_1, Olig_2, Olig_3, Olig_4), Astrocyte (Astro), Neuron (ExcL23, ExcL4, ExcL5_1, ExcL5_2, ExcL5_3, ExcL_6, Int_PV, Int_Sst_1, Int_Sst_2, Int_Vip, Int_Npy, Int_Cck), and Microglia (Micro_1, Micro_2). Single-nucleus RNA sequencing data were downloaded from GEO (GSE165578), reanalyzed, and visualized using Seurat (v3.2.2) as described (39). Briefly, cell types were annotated based on canonical markers: *PTPRC* for microglia, *PDGFRA* for OPCs, *MAG* for oligodendrocytes, *GFAP* and *SLC1A2* for astrocytes, and *CNTN5* and *NRG1* for neurons. In this study, the OPC subcluster of marmoset expressing high amounts of *BCAS1* and *ENPP6* was annotated as NFO.

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ACKNOWLEDGMENTS

We thank E. Muir for providing us with a plasmid containing a coding sequence of ChABC; M. Nathan for providing technical assistance with animal genotyping, tissue sectioning, and immunohistochemistry; C. Hawkins and other members of the JHU Transgenic Core for their assistance with the generation of transgenic mouse lines; A. Agarwal for his assistance with designing transgenic mouse lines; M. Pucak and A. Smirnov at the Neuroscience Imaging Core for their assistance with image acquisition and analysis; T. Babola for his assistance with data analysis; P. Parker for his assistance with cerebellar cranial windows; C. Call for his assistance with the graphics software; and L. Hsieh-Wilson and G. Zhang for helpful discussions about CSPG biochemistry. This research was supported (in part) by the Intramural Research Program of the National Institutes of Health (NIH). The contributions of the NIH authors were

made as part of their official duties as NIH federal employees, are in compliance with agency policy requirements, and are considered Works of the United States Government. However, the findings and conclusions presented in this paper are those of the authors and do not necessarily reflect the views of the NIH or the US Department of Health and Human Services. **Funding:** This work was funded by NIH grants AG072305 (D.E.B.) and NS041435 (D.E.B.); the Intramural Research Program of the National Institute of Neurological Disorders and Stroke (NINDS) (D.S.R.); the Dr. Miriam and Sheldon G. Adelson Medical Research Foundation (D.E.B., D.S.R.); and National MS Society no. FG-1708-28962 (Y.A.M.). **Author contributions:** Conceptualization: Y.A.M., D.E.B.; Methodology: Y.A.M., D.E.B., B.D., D.H., Y.K.T.X., A.Y.-H.H., J.E.v.B., G.C.M.-C., J.-P.L., D.S.R.; Investigation: Y.A.M., B.D., D.H., A.Y.-H.H., J.E.v.B., G.C.M.-C., A.A.K., J.-P.L.; Visualization: Y.A.M., D.H., J.-P.L., J.E.v.B.; Funding acquisition: D.E.B., Y.A.M., D.S.R.; Supervision: D.E.B., Y.A.M., D.S.R.; Software: Y.A.M., B.D., A.A.K., Y.K.T.X., D.H., J.-P.L.; Writing – original draft: Y.A.M., D.E.B. Writing – review & editing: Y.A.M., D.E.B., B.D., D.H., Y.K.T.X., A.Y.-H.H., J.E.v.B., G.C.M.-C., A.A.K., J.-P.L., D.S.R. **Competing interests:** The authors declare that they have no competing interests. **Data, code, and materials availability:** Data, code, and transgenic mouse genetic sequences used in the analysis are available on Dryad ([117](https://doi.org/10.5061/dryad.dr7sqvbc0)). **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adu2896

Figs. S1 to S12; Tables S1 to S6; MDAR Reproducibility Checklist; Movies S1 to S5

Submitted 31 October 2024; accepted 20 November 2025

[10.1126/science.adu2896](https://doi.org/10.1126/science.adu2896)



Myelin is repaired by constitutive differentiation of oligodendrocyte progenitors

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Science **391** (6783), eadu2896. DOI: 10.1126/science.adu2896

Editor's summary

Oligodendrocyte precursor cells (OPCs) are glial progenitor cells responsible for differentiating into oligodendrocytes that produce new myelin during learning (adaptive myelination) and also regenerate myelin lost through injury or disease. Mironova *et al.* studied OPC differentiation dynamics in mice during normal development and aging, as well as after demyelination and inflammation (see the Perspective by Macklin and Allen). The authors showed that OPCs attempt to differentiate continuously throughout the central nervous system at a constant rate independently of myelin demand. Moreover, demyelination did not alter the OPC differentiation rate, indicating that OPC differentiation is cell autonomous. —Mattia Maroso

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